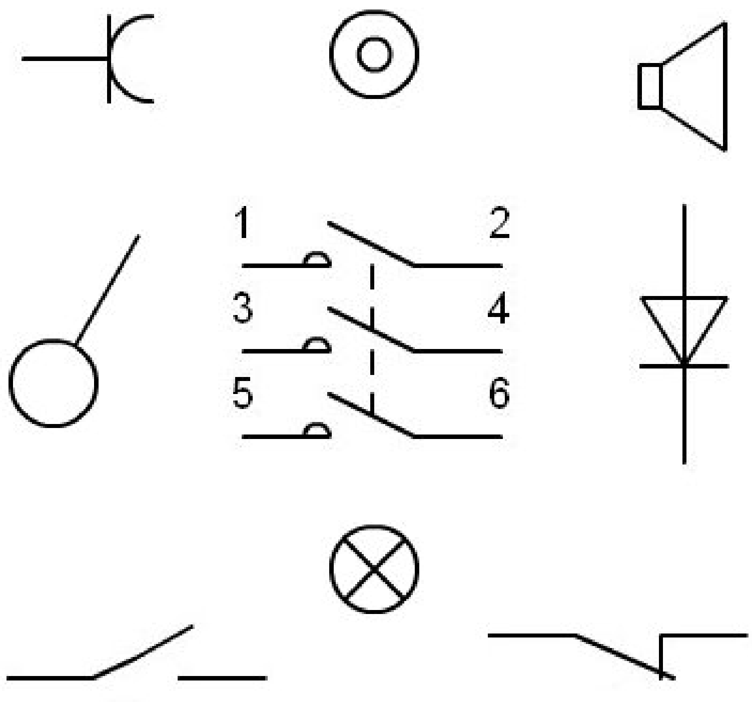
Autor:Antonio Bueno

Teaching unit:
Electrical Symbolism



3rd ESO Course

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Didactic unit:
Electrical Symbolism

INDEX

1.- Standard UNE-EN 60617 (IEC 60617)

In recent years (1996 to 1999), the graphic symbols for diagrams have been modified. electrical, at the international level with the standard IEC 60617, which has been adopted at the European level in the standard EN 60617 which has finally been published in Spain as the standard UNE-EN 60617.

It is necessary to make known the most used symbols. The consultation of these symbols by computer media in the competent bodies that publish them (CENELEC and others) is subject to subscription and payment, so I have deemed it appropriate to publish this commented excerpt, where I can consult some of the most common symbols for free.

This standard is divided into the following parts:

Part	Description
IEC 60617-2	Elements of symbols, distinctive symbols, and other symbols of general application
UNE-EN 60617-3	**Conductors and connecting devices**
UNE-EN 60617-4	**Basic Passive Components**
UNE-EN 60617-5	**Semiconductors and electronic tubes**
UNE-EN 60617-6	**Production, transformation, and conversion of electrical energy**
UNE-EN 60617-7	**Apparatus and control and protection devices**
IEC 60617-8	Measuring instruments, lamps and devices of signaling
UNE-EN 60617-9	**Telecommunications: Switching and peripheral equipment**
UNE-EN 60617-10	Telecommunications: Transmission
UNE-EN 60617-11	Installation diagrams and plans, architectural and topographical.
UNE-EN 60617-12	**Binary logical operators**
UNE-EN 60617-13	**Analog operators**

To know all the symbols in detail, as well as the representation of new symbols, one must consult the full regulation.

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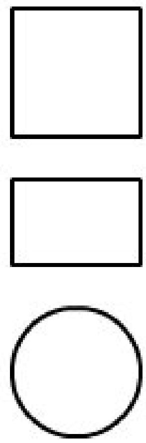
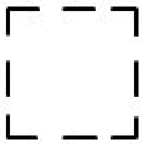
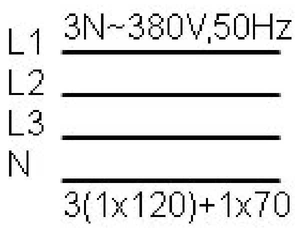
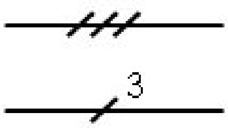
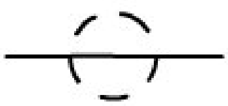
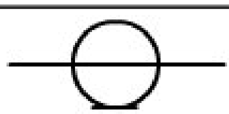
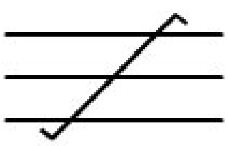
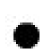
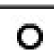
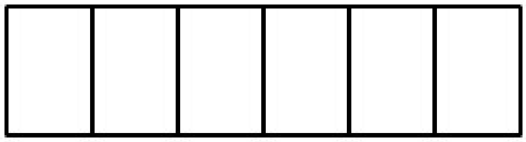
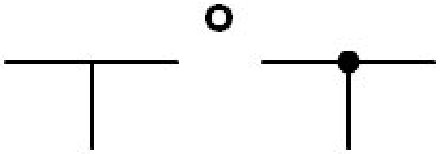
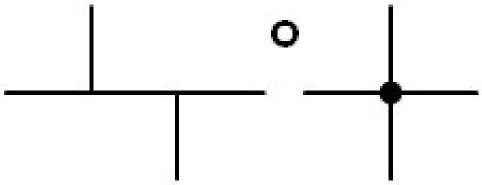
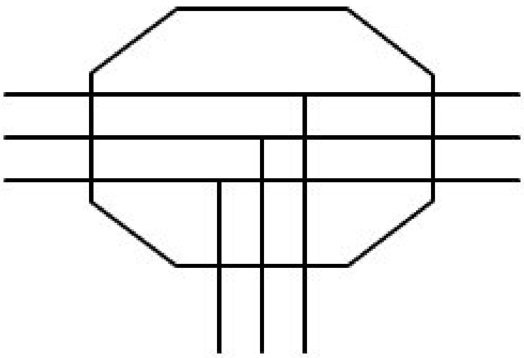
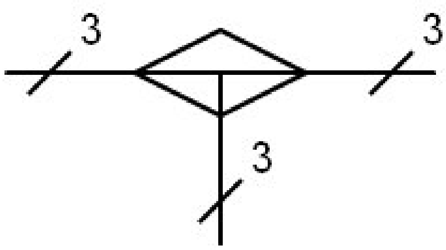
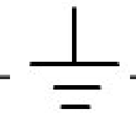
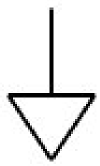
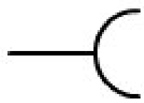
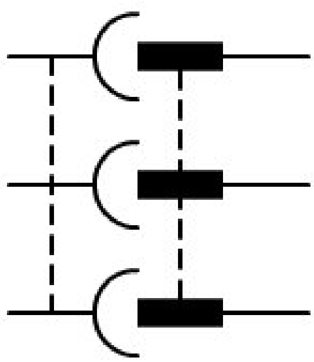
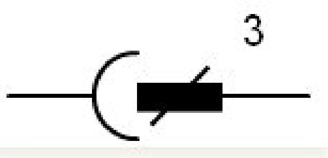
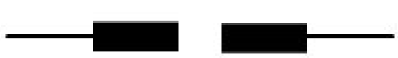
2.- Conductors, passive components, control elements and basic protection

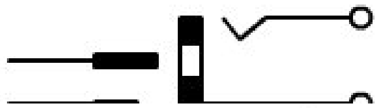
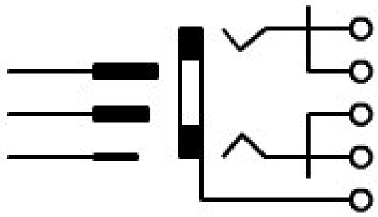
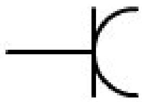
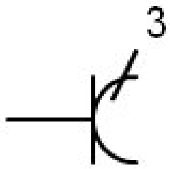
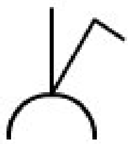
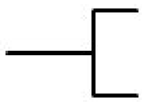
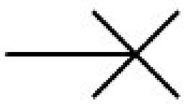
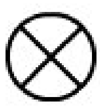
The most commonly used symbols in electrical installations are the following:

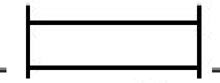
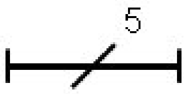
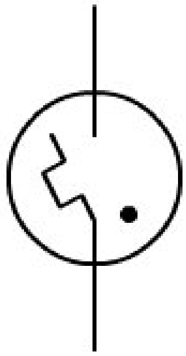
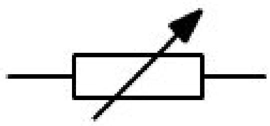
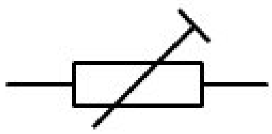
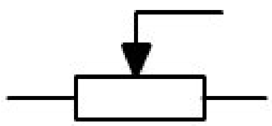
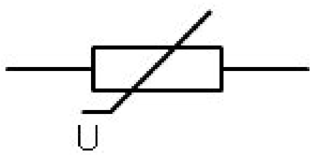
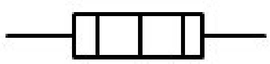
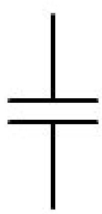
Symbol	Description
	Object (contour of an Object)

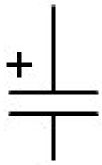
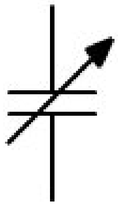
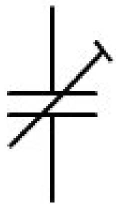
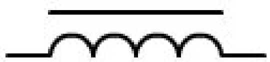
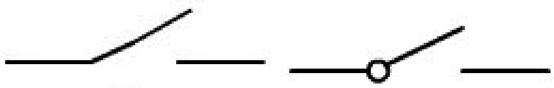
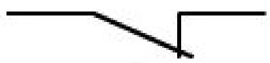
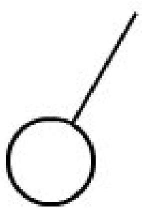
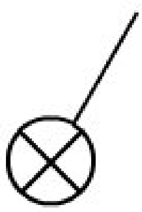
	For example: Team Device Functional unit - Component Function They must be incorporated into the symbol or placed in their proximity other symbols or appropriate descriptions to specify the type of object. If representation requires it, it can be done. use a contour in another way
	**Screen, Shielding** For example, to reduce penetration of fields electric o electromagnetic. The symbol must be drawn with the shape that is convenient.
	Conductor
	**Conductor** If can to give information complementary. Example: three-phase current circuit, 380 V, 50 Hz, three conductors of 120 mm^2, with neutral thread of 70 mm^2
	**Conductors (single-wire)** Both representations are correct. Example: 3 drivers
	**Flexible connection**
	**Shielded conductor**
	**coaxial cable**
	**Twisted connection** 3 connections are shown
	**Union** Connection point
	**Terminal**

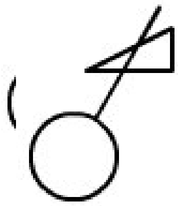
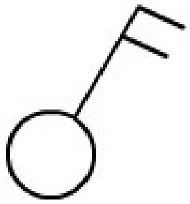
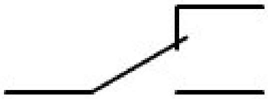
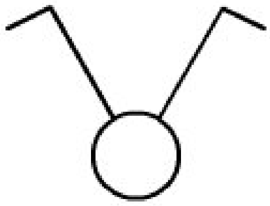
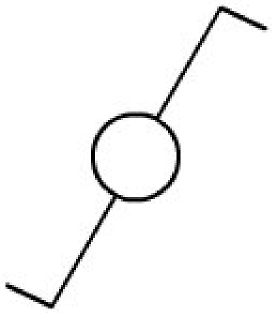
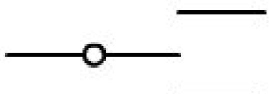
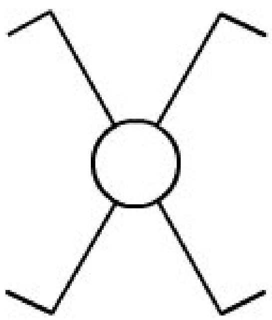
	**Terminal strip** Terminal marks can be added.
	**Connection in T**
	**Double union of conductors** Form 2 should only be used if is necessary by reasons of representation.
	Junction box, shown with three conductors with T connections. Multilinear representation.
	Junction box, shown with three conductors with T connections. Single line representation.
	Direct current
	**Alternating current**
	**Current rectified with alternate component.** (If it is necessary to distinguish it from a) rectified and filtered current
+	**Positive polarity**
—	**Negative polarity**
N	**Neutral**
	Earth

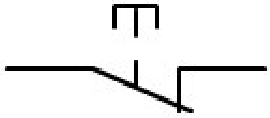
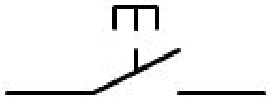
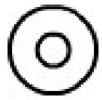
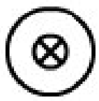
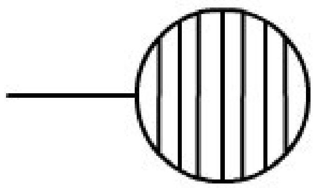
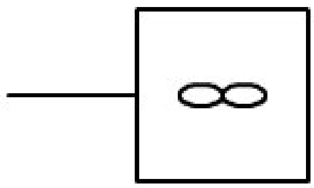
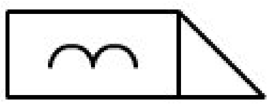
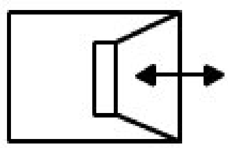
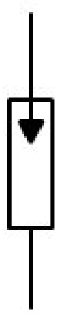
	Additional information can be provided about the state of the earth if its purpose is not evident.
	**Masa, Chassis** It can be omitted completely or partially. the stripes if there is no ambiguity. If they omit, the mass line must be more thick.
	**Equipotentiality**
	Female connector (from a base or of a plug). Socket base. In a single line representation, the symbol indicates the female part of a connector multicontact.
	Male contact (of a base or of a plug). Plug. In a single-line representation, the symbol indicates the male part of a connector multicontact.
	**Base and Plug**
	**Base and multipole connectors** The symbol is displayed in a multifilar representation with 3 contacts female and 3 male connectors.
	**Base and multipolar connector** The symbol is displayed in a single-line representation with 3 contacts female and 3 male contacts.
	**Push connector**

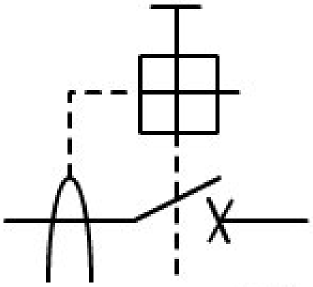
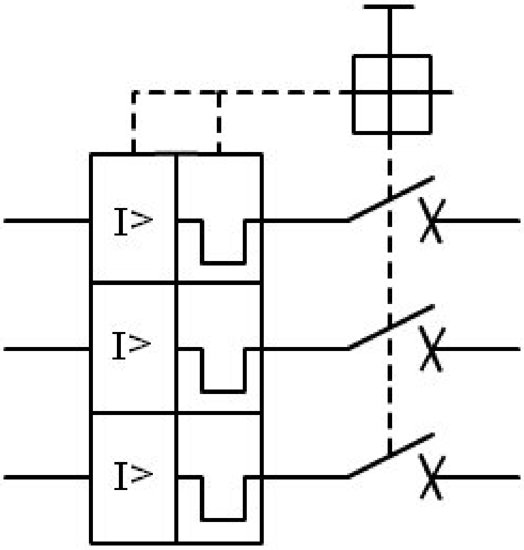
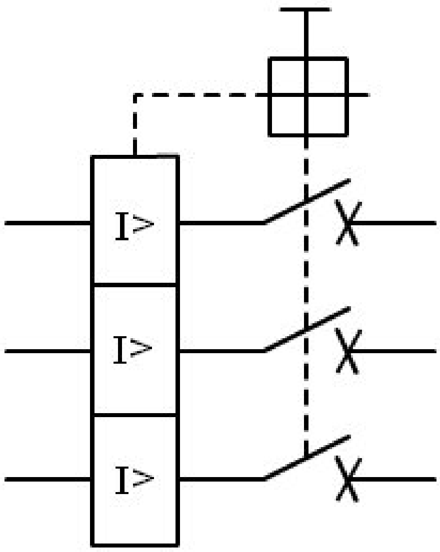
	**Plug and jack connector**
	**Plug and jack connector with breakup contacts**
	**Base with contact for driver of protection**
	**Multiple outlet plug** The symbol represents 3 contacts. female with protective conductor
	**Socket base with switch unipolar**
	**Base of plug (telecommunications)** Symbol general. The designations can be used to distinguish different types of shots: TP = teléfono FX = telefax M = micrófono FM = frequency modulation TV = televisión TX = telex speaker
	**Starting point for apparatus of lighting** Symbol represented with wiring.
	Lamp, general symbol.
	Luminaria, general symbol. **Lamp fluorescent** symbol general.
	Luminaria with three tubes

	**fluorescent (multi-wire)**
	**Lantern with five tubes fluorescent (single wire)**
	**Brewer, Gas discharge tube with** Starter thermal for lamp fluorescent.
	**Resistance, general symbol.**
	**Photoresistor**
	**Variable resistor**
	**Resistance variable of value preset**
	**Potentiometer with movable contact**
	**Resistance dependent of the tension**
	**Heating element**
	Capacitor, general symbol.

	**Polarized capacitor, capacitor** electrolytic.
	**Variable capacitor**
	**Condenser with adjustment default**
	Coil, general symbol, inductance, **winding or reactance**
	**Coil with magnetic core**
	**Reel with fixed connections, it is shown a** intermediate intake.
	**Normally Open Switch (NO).** Either of the two symbols is valid.
	**Switch normally closed (NC).**
	**Switch automatic.** Symbol general.
	**Switch. Single-line.**
	**Switch with pilot light. Single wire.**
	Single-pole switch with time of limited connection. Single wire.

	
	**Dimmer switch. Single wire.** Light intensity regulator.
	**Bipolar switch. Single wire.**
	**Switch**
	**Unipolar switch. Single wire.** For example, for the different levels of lighting.
	**Switch unipolar of dos positions. Toggle switch.** Single wire.
	**Switch with positioning cutting intermediate.**
	**Intermediate switch. Switch cross. Single wire.** Equivalent circuit diagram.

	**Normally closed button**
	**Normally open push button**
	**Push button. Single wire.**
	**Button with indicator light.** Single-phase.
	**Heater of water.** Symbol represented with wiring.
	**Fan. Symbol represented with** wiring.
	**Electric lock**
	**Intercom.** For example: intercom.
	**Fusible**
	**Fuse Switch**
	**Lightning rod**

	**Differential circuit breaker.** Represented by two poles.
	**Switch automatic circuit breaker or motor protector.** Represented by three poles.
	**maximum circuit breaker intensity. Circuit breaker magnetic.**

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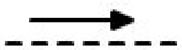
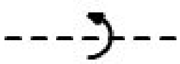
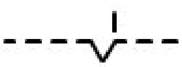
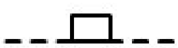
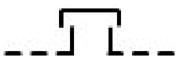
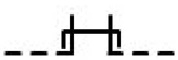
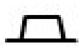
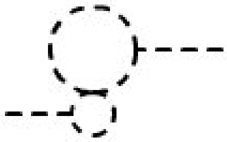
3.- Power switching devices, relays, contacts and actions

The obtaining of the different symbols is formed from the combination of couplings. actuators and other basic symbols. Below are the most important ones and then some of the most common symbols.

Mechanical couplings	
Symbol	**Description**
	Connection, mechanical, hydraulic, optical or functional. The length can be adjusted as needed.
	Connection, mechanical, hydraulic, optical or functional. It is only used when the previous form cannot be used.
	Connection, indicating the direction of force or movement of the translation.
	Connection, indicating the direction of the movement of the rotation.
	**Delayed action.** Form 1 and form 2
	**With automatic return.** The triangle is heading towards the direction of return.
	**Ratchet, non-return or non-automatic return.** Device to maintain a given position.
	**Released strut or stop**
	**Wedge or fitted stop**
	**Mechanical interlocking between two devices**
	**Release latch device**
	**Coupling device hooked**
	**Locking device**
	**Mechanical clutch disengaged**
	**Mechanical clutch engaged**
	**Brake**

	**Gear**

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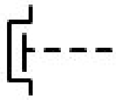
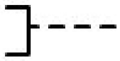
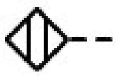
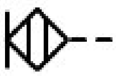
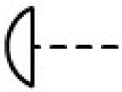
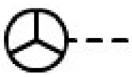
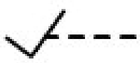
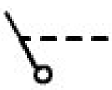
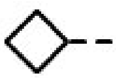
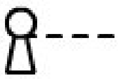
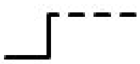
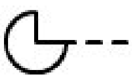
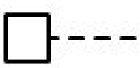
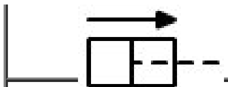
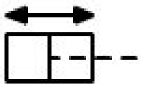
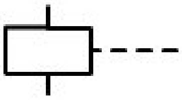
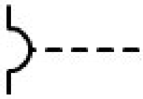
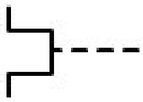
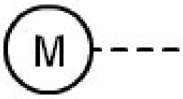
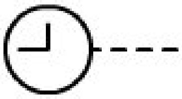
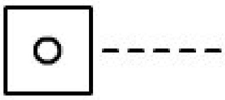
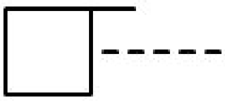
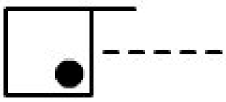
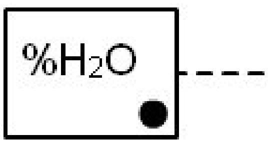
Device actuators	
Symbol	**Description**
	**Manual actuator, general symbol**
	**Manual actuator protected against unauthorized operation** intentional. Button with protective housing protection against unauthorized manipulation
	**Shooter's controller. Shooters**
	Rotary dial. Selectors, switches.
	**Push button remote. Push buttons**
	**Remote control by proximity effect. Inductive sensors of** proximity.
	**Contact command. Probes**
	**Emergency drive type "mushroom". Push button of** emergency stop
	**Steering wheel control.**
	**Pedal remote.**
	**Lever control.**
	**Detachable manual control.**
	**Key remote.**
	**Crank handle.**
	**Slide or pulley control. Limit switch**
	Mando de leva. Interruptor de leva
	**Sending by energy accumulation.**

	**Hydraulic or pneumatic drive of simple effect.**
	**Drive by hydraulic or pneumatic energy, of double effect.**
	**Electromagnetic effect drive. Relay.**
	**Actuation by an electromagnetic device for** overcurrent protection
	**Actuation by a thermal device for protection** against overintensity
	**Command by electric motor**
	**Sending by electric clock**
	**Fluid level actuation. Level float** water
	**Driven by a counter. Counts impulses**
	**Activated by the flow of a fluid. Flow switch.** water
	**Driven by the flow of a gas. Air flow switch**
	**Activated by relative humidity.**

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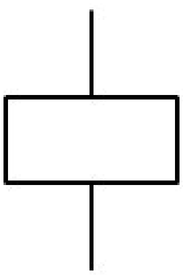
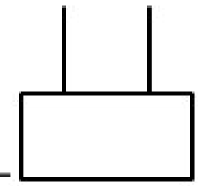
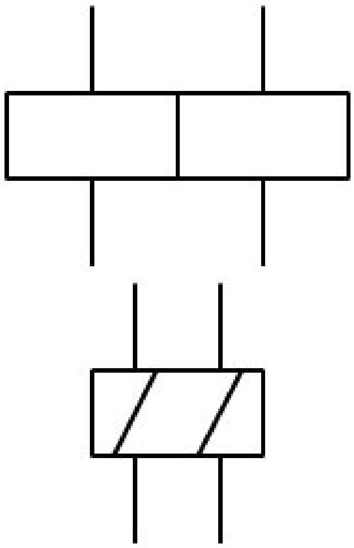
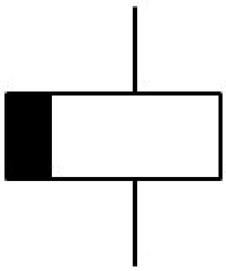
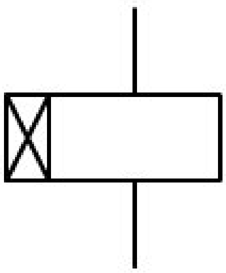
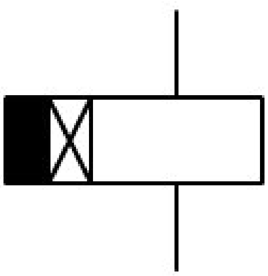
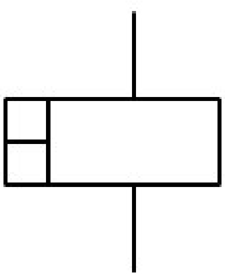
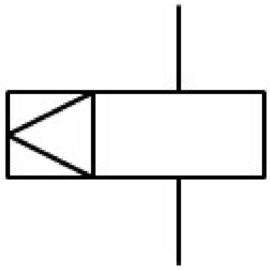
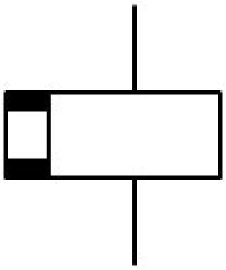
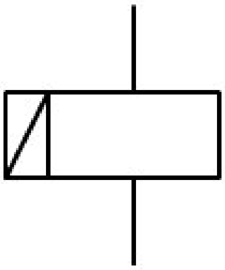
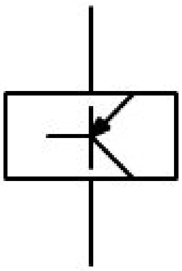
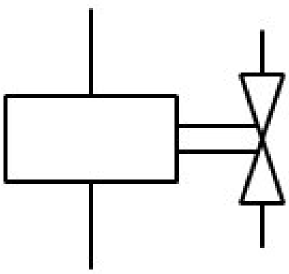
Relays	
Symbol	**Description**
	**Relay coil, contactor or other device** command, general symbol. Either of the two symbols is valid. If a device has several windings, it can be

	indicate by adding the number of diagonal strokes in the interior of the symbol.
	Example: Control device with two windings separated. Form 1 and form 2
	**Delayed disconnection control device.** Delayed disconnection when activating the remote.
	**Delayed control device upon connection.** Delayed connection when activating the remote.
	**Delayed control device for the connection and to** the disconnection. Delayed connection when activating the command and also when deactivating it.
	**Quick relay remote. Connection and disconnection** fast (special relays).

	**Control of a mechanical interlocking relay.** Relay
	**Control of a polarized relay.**
	**Control of a latching relay.**
	**Electronic relay remote.**
	**Solenoid of a solenoid valve.**

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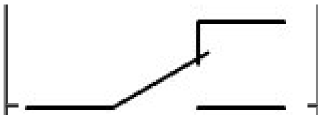
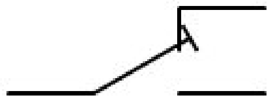
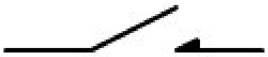
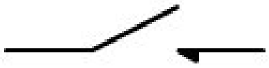
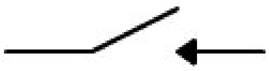
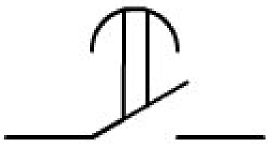
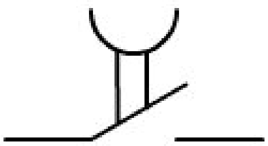
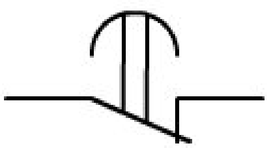
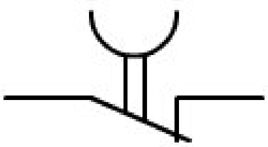
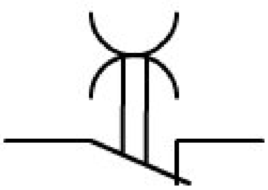
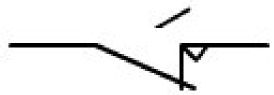
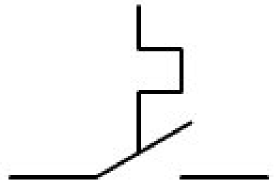
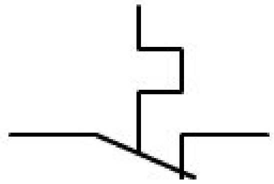
Control element contacts	
Symbol	**Description**
	**Normally open switch (NO).**
	**Normally closed switch (NC).**

	**Switch.**
	**Overlapping investor contact. Close the NO before opening.** NC
	**Temporary contact, with momentary closure when its control device is activated.**
	**Pass contact, with temporary closure when its control device is deactivated.**
	**Instant contact, with momentary closure when its The control device is activated or deactivated.**
	Contact (from a set of several contacts) for closure **advanced compared to other contacts of set.**
	Contact (of a set of several contacts) for closure **delayed compared to the other contacts in the set.**
	Opening contact (of a set of several contacts) **delayed compared to the other contacts in the group.**
	Opening contact (of a set of several contacts) **ahead of the other contacts of set.**
	**Delayed closure contact to the connection of your** Control device. Connection timer
	**Delayed closure contact to disconnection of your** control device. Timer for disconnection
	**Delayed opening contact to the connection of your** control device. Connection timer
	**Delayed opening contact to the disconnection of your** control device. Timer for disconnection
	**Delayed closing contact to the connection and also to the disconnection of your control device.**
	Automatic return closing contact.

	**Opening contact with automatic return.**
	**Auxiliary contact of self-acting closure by a relay thermal.**
	**Auxiliary contact for relay-operated self-opening thermal.**

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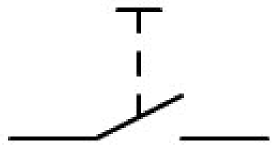
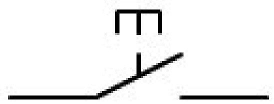
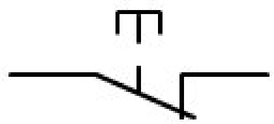
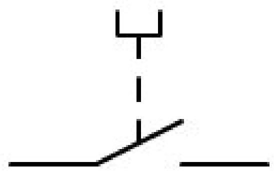
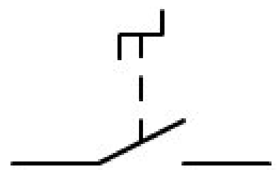
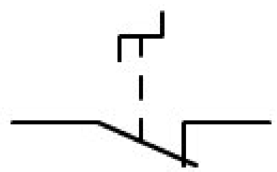
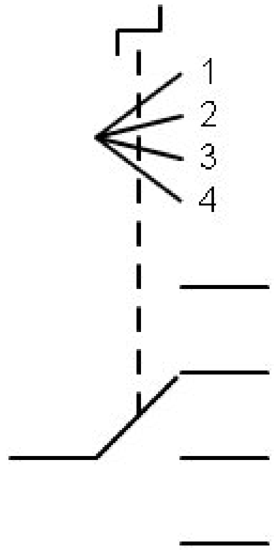
Contacts of manual control actuators	
Symbol	**Description**
	**Manual control closure contact, general symbol** Control switch
	**Normally open button. (automatic return)**
	**Normally closed button (automatic return)**
	**Rotary switch.**
	**Turn switch with closing contact.**
	**Turn switch with opening contact.**

	Example of a 4-position rotary control switch fixed

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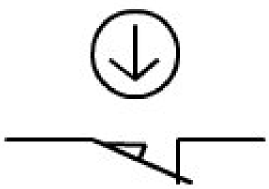
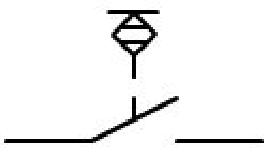
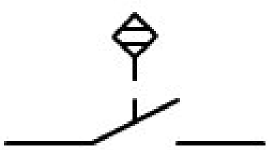
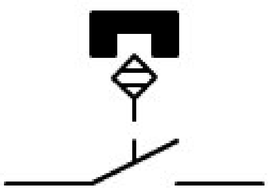
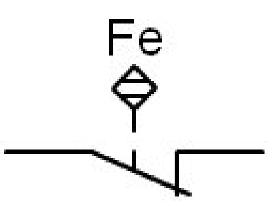
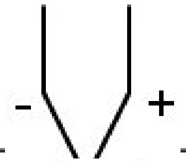
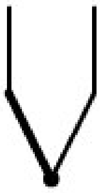
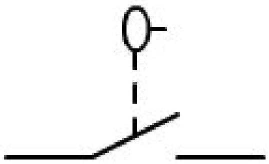
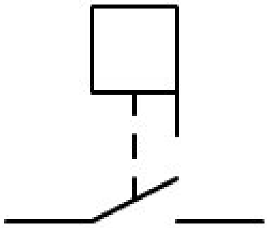
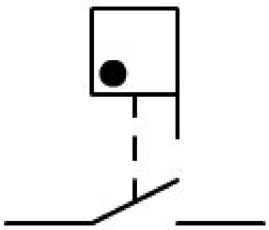
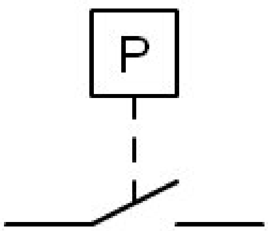
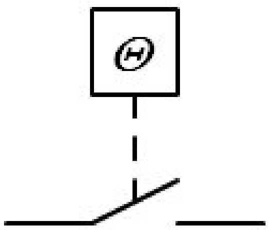
Field capturing elements	
Symbol	**Description**
	**Position switch closing contact.** Contact NO of a degree completion
	**Opening contact of a position switch.** NC contact of a graduation finish
	**Opening contact of a position switch with positive opening maneuver. End of the race of** security.
	**Touch-sensitive switch with normally closed contact.**
	**Proximity switch with normally closed contact.** Inductive sensor for metallic materials
	**Proximity switch with normally closed contact driven by magnet.**
	**Proximity switch for ferrous materials with opening contact.** Iron (Fe) proximity detector

	**Thermocouple, represented with polarity symbols.**
	**The polarity of the thermocouple is indicated with the thicker line in** one of its terminals (negative pole)
	**Fluid level switch.**
	**Fluid flow switch**
	**Gas flow switch**
	**Pressure switch (pressure stat)**
	**Temperature activated switch (thermostat)**

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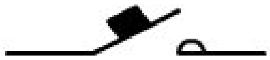
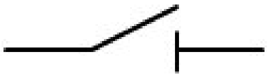
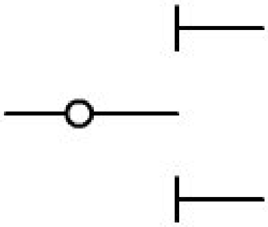
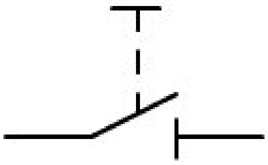
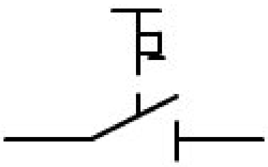
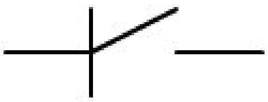
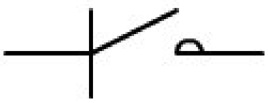
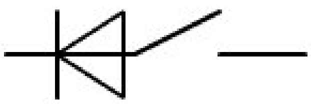
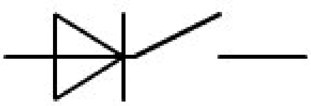
Power elements	
Symbol	**Description**
	Contactor, main closing contact of a

	contactor. Open contact in rest.
	**Contactor, main opening contact of a** contactor. Contact closed at rest.
	**Contactor with automatic disconnection triggered by a measuring relay or an incorporated trigger.**
	**Disconnecting switch.**
	**Two-position switch with position intermediate**
	**Disconnector (manual control)**
	**Automatic opening disconnect switch triggered by a measurement relay or a trigger incorporated**
	**Isolating switch (manual control) Switch** section switch with locking device
	Static switch, (semiconductor) general symbol.
	**Static contactor, (semiconductor).**
	Static contactor, (semiconductor) with the passage of the current in one direction. Left.
	Static contactor, (semiconductor) with the passage of the current in one direction. Rights.

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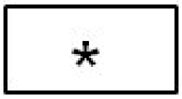
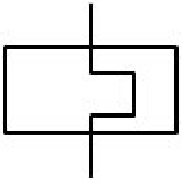
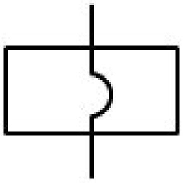
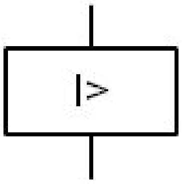
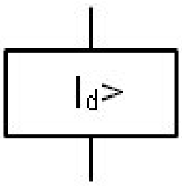
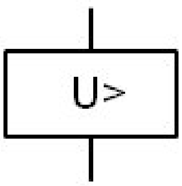
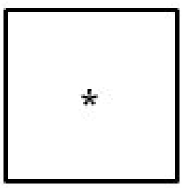
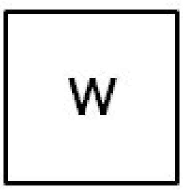
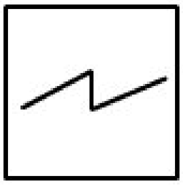
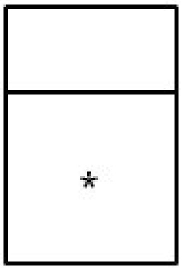
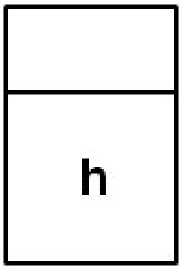
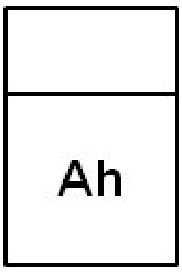
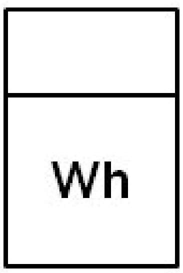
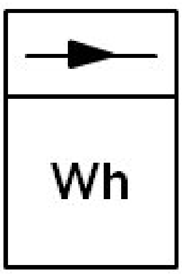
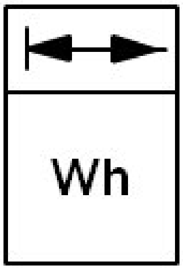
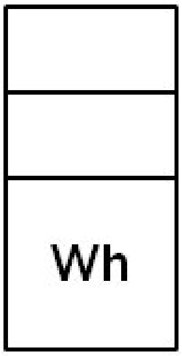
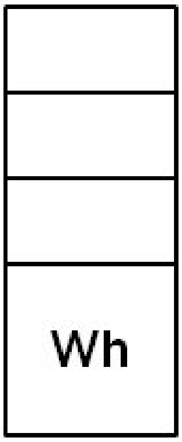
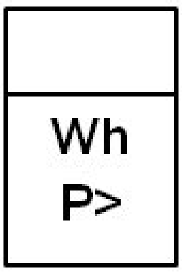
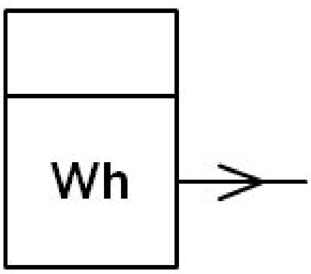
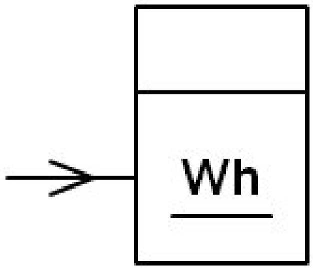
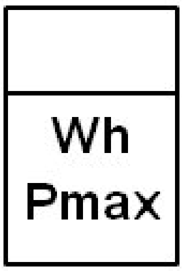
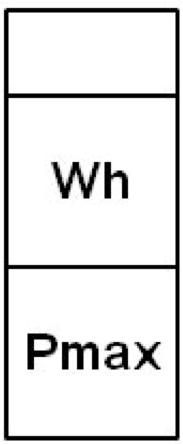
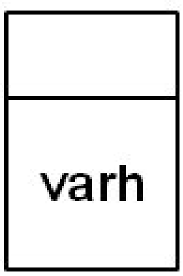
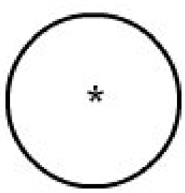
4.- Measuring and signaling instruments

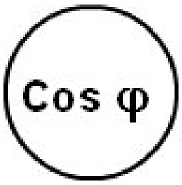
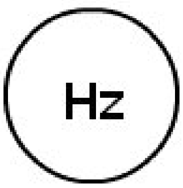
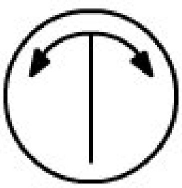
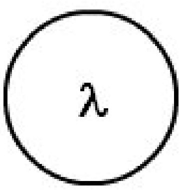
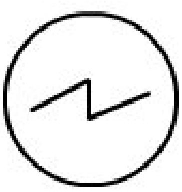
Symbol	Description

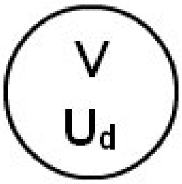
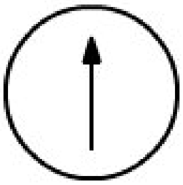
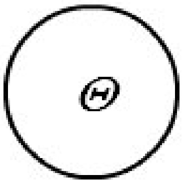
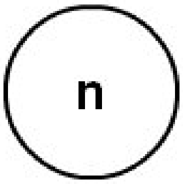
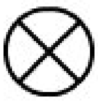
	**Measurement relay.** Device related to a measuring relay. 1.- The asterisk should be replaced by one or more letters or distinctive symbols that indicate the parameters of the device in the following order: Characteristic magnitude and its variation form. Sense of the flow of energy. Adjustment field. Restoration relationship. Delayed action. Value of time delay
	**Thermal relay.**
	**Electromagnetic relay.**
	**Maximum current relay (overcurrent)**
	**Differential current relay (Id)**
	**Overvoltage relay**
	**Recording device. General symbol.** The asterisk is replaced by the symbol of magnitude that will register the device
	**Wattmeter recorder.**

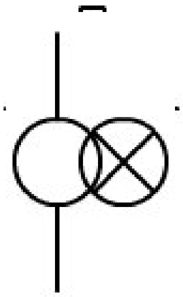
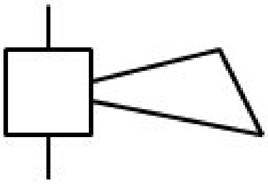
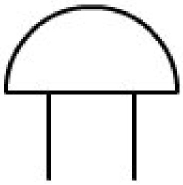
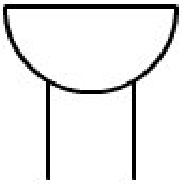
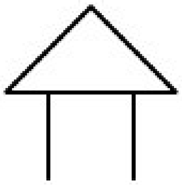
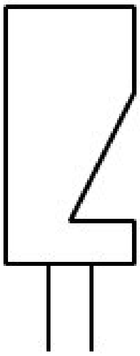
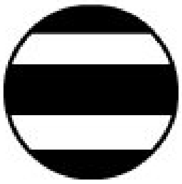
	**Oscilloscope.**
	**Integrating apparatus. General symbol.** The asterisk is replaced by the measurement magnitude.
	Hourly counter. Hour counter.
	**Amperemeter. Ampere-hour meter.**
	**Active energy meter. Varihorimeter. Meter** of watt-hours
	**Active energy meter, which measures energy one-way transmission. Watt-hour counter**

	**Energy exchanged meter (to and from)** bars Watt-hour meter
	**Dual tariff active energy meter**
	**Triple rate active energy meter**
	**Excess active power energy meter**
	**Active energy meter with data transmitter**
	Repeated from an active energy meter

	
	**Repeated from an active energy meter with a printing device**
	**Active energy meter with value indication maximum of the average power**
	**Active energy meter with value recorder maximum of the average power**
	**Reactive energy meter. Variometer. Meter** reactive volt-amperes per hour
	**Indicator device. General symbol.** The asterisk is replaced by the symbol of the magnitude that It will indicate the device. Examples: A = Ammeter. mA = milliampere. V = Voltmeter. W = Wattmeter.

	Voltmeter. Voltage indicator.
	**Reactive current ammeter.**
	Várometer. Reactive power indicator.
	**Power factor measuring device.**
	Phase meter. Phase angle indicator.
	Frequency meter. Frequency indicator.
	Synchronoscope. Indicator of the phase difference between two signals. for your synchronization.
	Wavemeter. Indicator of the wavelength.
	Oscilloscope. Waveform indicator.

	Differential voltmeter. Indicator of the difference of tension between two signals.
	Galvanometer. Indicator of galvanic isolation.
	**Thermometer. Pyrometer. Temperature indicator.**
	Tachometer. Indicator of revolutions.
	**Signal lamp, general symbol.** Si se desea indicar el color, se debe colocar el siguiente code next to the symbol: RD or C2 = red OG or C3 = Orange YE or C4 = yellow GN or C5 = green BU or C6 = blue WH ó C9 = white If you want to indicate the type of lamp, you must place the next code along with the symbol: Ne = neón Xe = xenón Na = sodium vapor Hg = mercurio I = iodine incandescent EL = electromagnetic ARC = arco FL = fluorescent IR = infrared UV = [illegible] LED = light-emitting diode.
	Oscillating signal lamp.

	**Lamp fed through transformer incorporated.**
	**speaker**
	**Timbre, bell**
	**Buzzard**
	**Siren**
	**Electric drive whistle**
	**Electromechanical signaling element**

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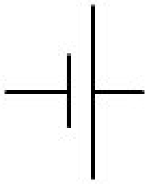
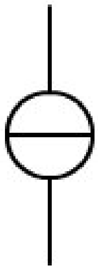
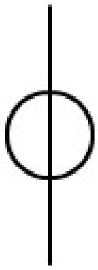
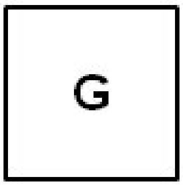
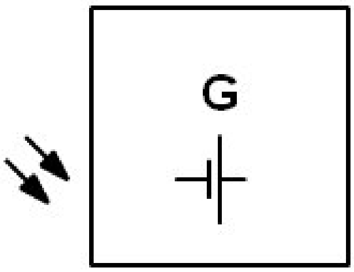
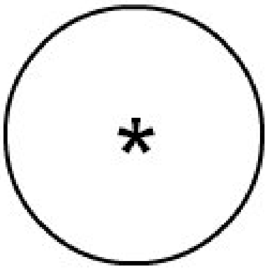
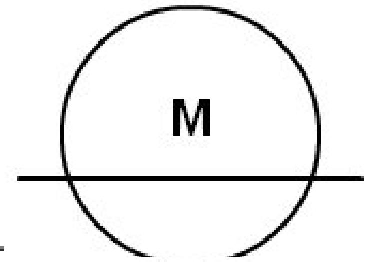
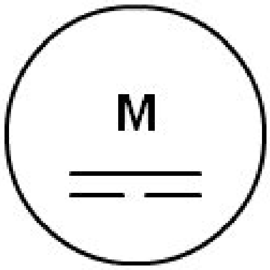
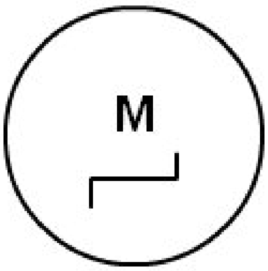
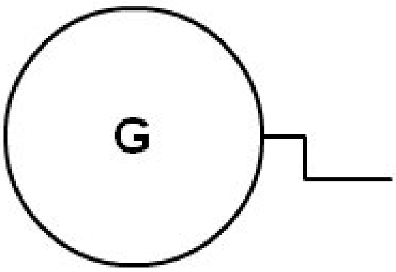
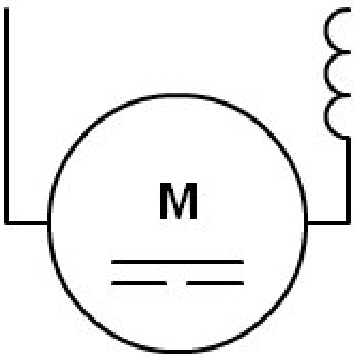
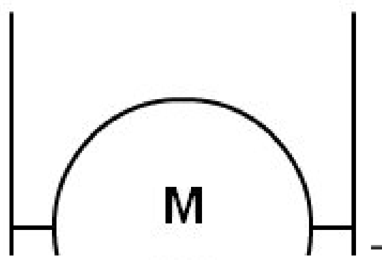
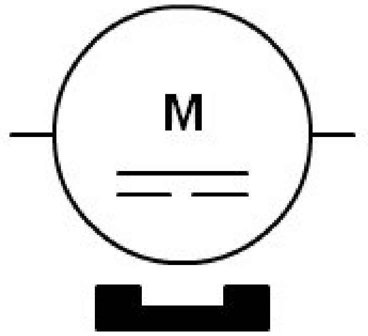
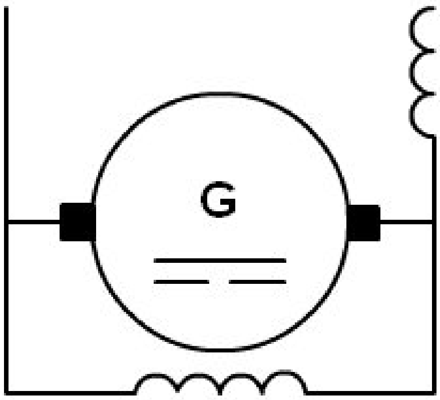
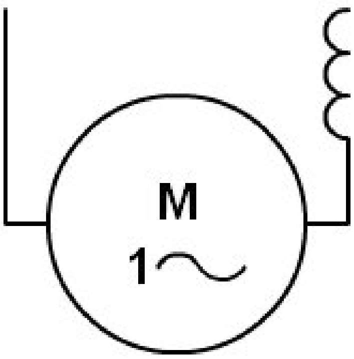
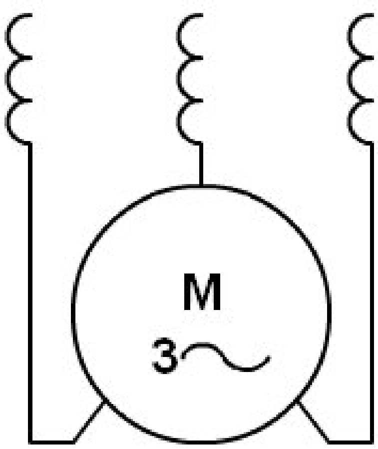
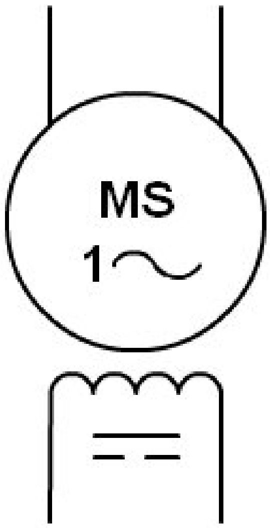
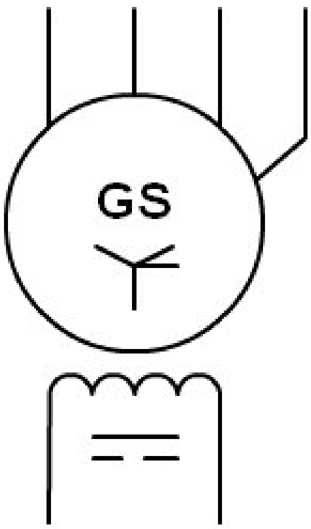
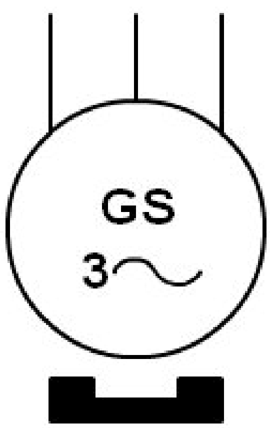
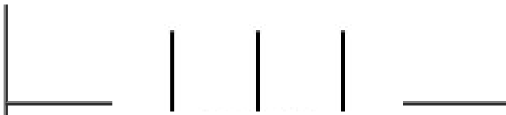
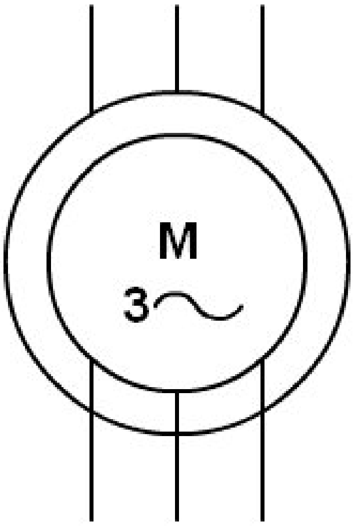
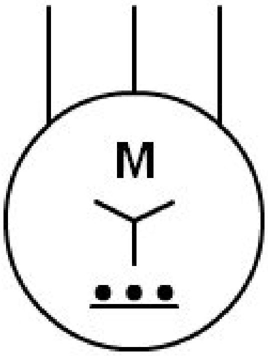
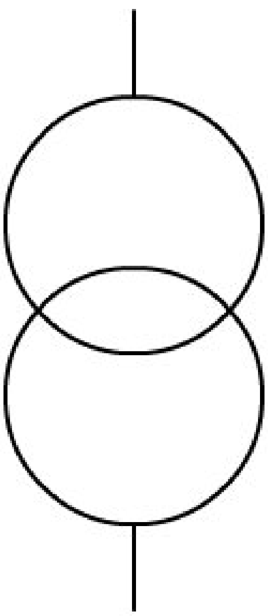
5.- Production, transformation and conversion of electrical energy

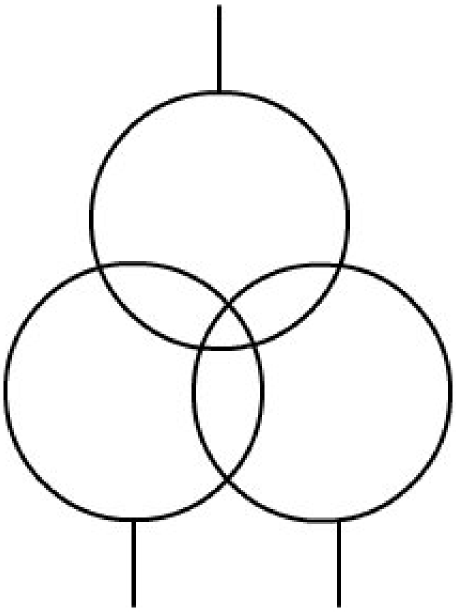
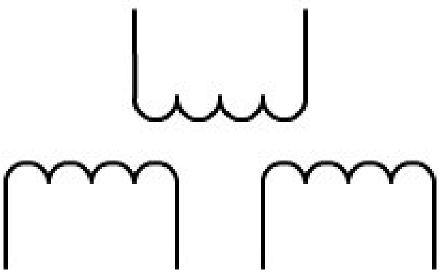
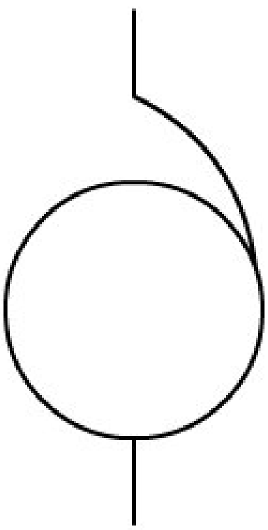
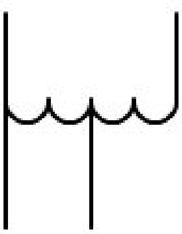
Symbol	Description
	Battery or accumulator, the long stroke indicates the positive
	**Ideal current source.**
	**Ideal voltage source.**
	**Non-rotating generator. General symbol**
	**Photovoltaic generator**
	**Rotary machine. General symbol.** The asterisk, *, will be replaced by one of the following literal symbols: **C= Switch matrix** **G= Generador** **GS= Synchronous generator** **M= Motor** MG=Reversible machine (that can be used as a motor and generator MS=Synchronous motor

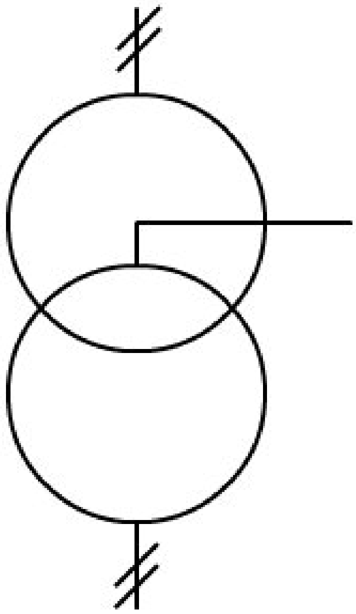
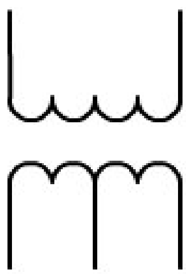
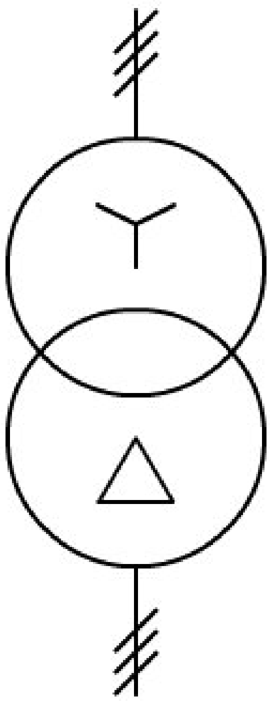
	Linear motor. General symbol.
	**Direct current motor.**
	**Stepper motor.**
	Manual generator. Electric generator of calling, magnet.
	Series motor, direct current
	Shunt excitation motor direct current

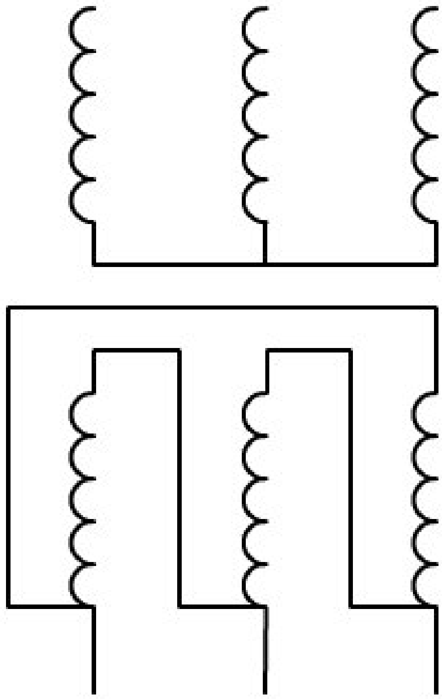
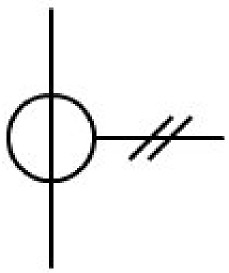
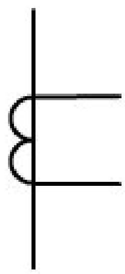
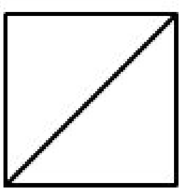
	
	**Permanent magnet direct current motor permanent.**
	**Direct current generator with short composite excitation, represented** with terminals and brushes.
	**Single-phase series collector motor.** Alternating current machine.
	**Three-phase motor. Collector machine.**

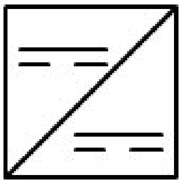
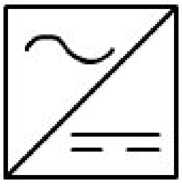
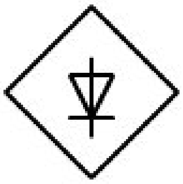
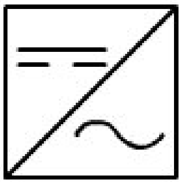
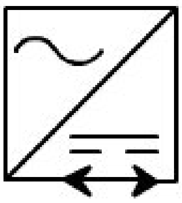
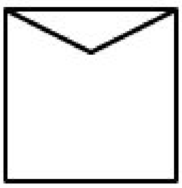
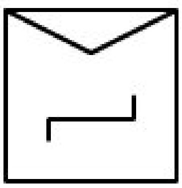
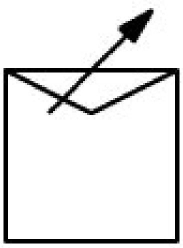
	**Single-phase synchronous motor.**
	**Three-phase synchronous generator, with rotor in an accessible star and neutral.**
	**Three-phase synchronous generator with magnets permanent.**
	Three-phase induction motor with rotor in squirrel cage.

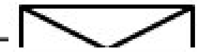
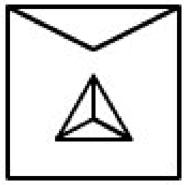
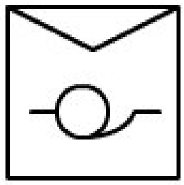
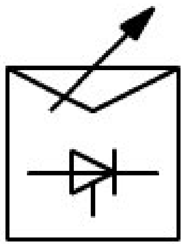
	
	**Three-phase induction motor with rotor winding.**
	**Three-phase induction motor with stator in a star and automatic starter incorporated.**
	**Two winding transformer (single-phase). Single wire**
	Two winding transformer (single-phase). Multi-wire

	
	**Transformer with three windings.** Single wire
	**Three winding transformer.** Multifilar
	**Autotransformer. Single-line**
	**Autotransformer. Multicore**

	**Transformer with midpoint tap in a wiring. Single line**
	**Transformer with intermediate tap in a winding.Multi-wire**
	**Three-phase transformer, star connection - triangular. Single wire**

	**Three-phase transformer, star connection - triangular.Multiwire**
	**Transformer of current o impulse transformer. Single line**
	**Transformer of current o pulse transformer. Multi-wire**
	Converter.General symbol. Both sides of the bar can be indicated. central a symbol of magnitude, shape of wave, etc. for input and output to indicate the nature of conversion.

	**DC/DC converter.**
	**Rectifier. General symbol (converter of** AC and DC
	**Double wave rectifier (bridge** rectifier.
	**Charger, Inverter (DC to AC converter).** AC)
	**Rectifier / Modulator; Rectifier /** **inverter.**
	**Motor starter. General symbol.** Single-phase.
	**Step motor starter. It can be.** indicate the number of stages. Single wire.
	**Starter regulator Variable speed drive** **speed.Single wire.**
	Direct starter with contactors for **change the direction of rotation of the motor.** Single-wire.

	
	**Starters - triangle. Single line.**
	**Auto-transformer starter. Single line.**
	**Starter - thyristor regulator, Frequency converters, Inverters of speed. Unifilar.**

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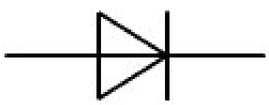
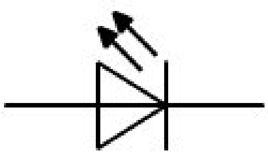
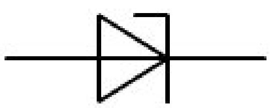
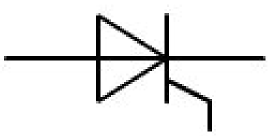
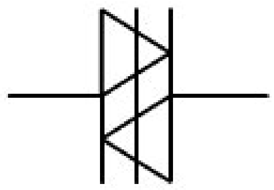
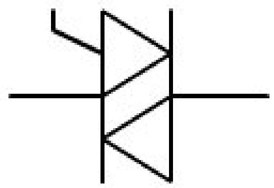
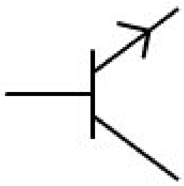
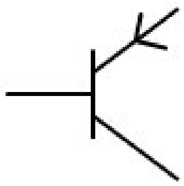
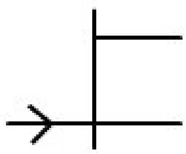
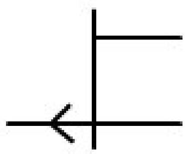
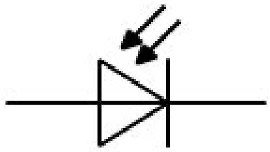
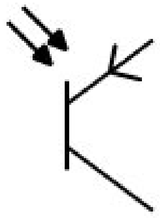
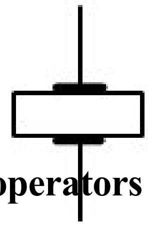
6.- Semiconductors

Symbol	Description
	**Diode**
	**Light Emitting Diode (LED)**
	**Zener diode**
	**Thyristor**

	**Diac. Bidirectional thyristor diode.**
	**Triac. Bidirectional thyristor triode.**
	**NPN bipolar transistor**
	**Bipolar PNP transistor**
	**N-channel field-effect transistor (FET)**
	**P-type channel field effect transistor (FET)**
	**Photodiode**
	**Phototransistor**
	Piezoelectric crystal



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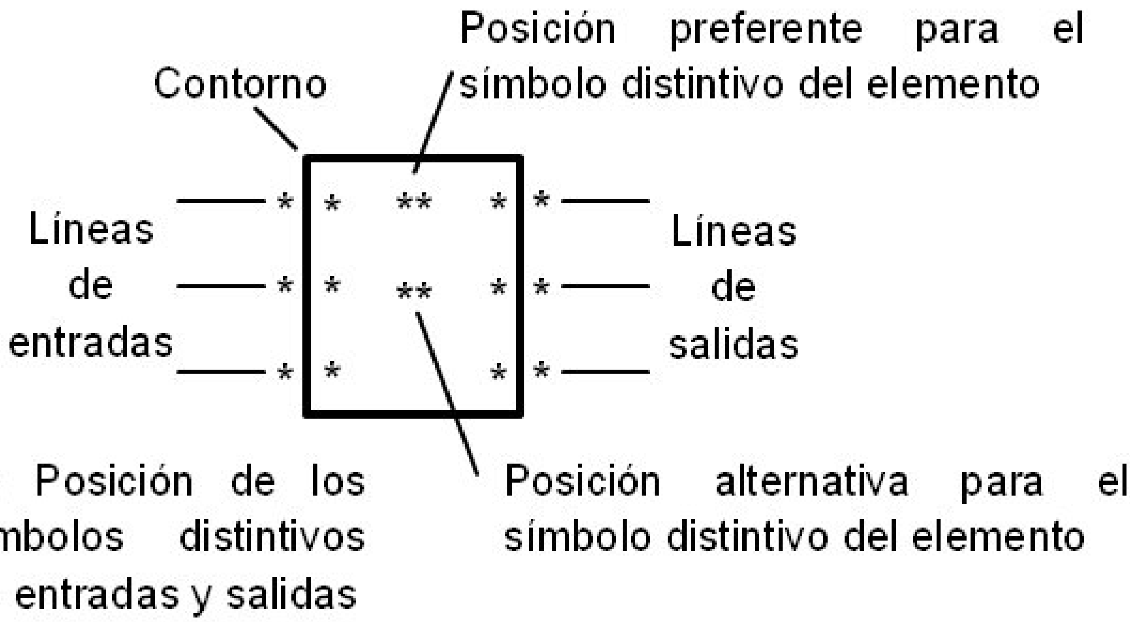
7.- Analog operators

Given the complexity that these symbols can have, they will be composed of the following parts:

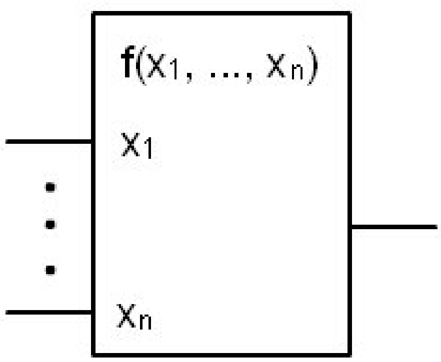
Contour or set of contours, along with one or more distinctive symbols and the entry lines and of exit.

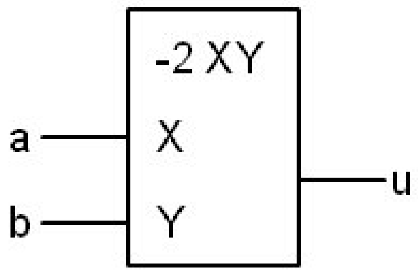
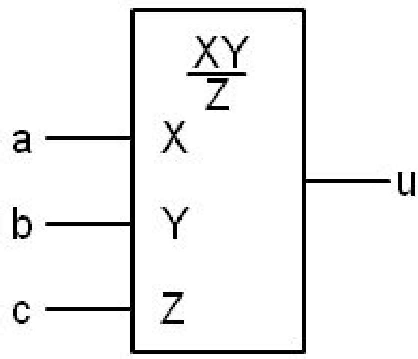
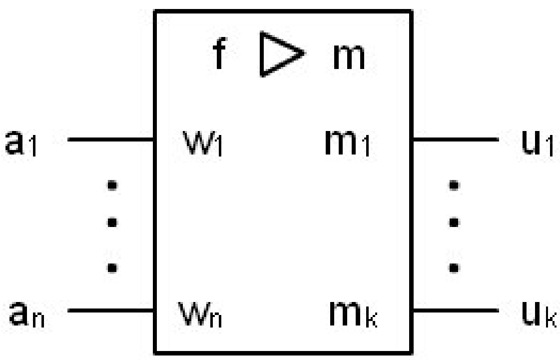
The basic scheme of this symbol is:

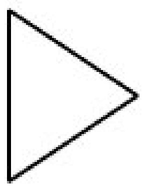
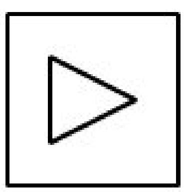


The relationship between the width and length of the perimeter is arbitrary.

Unless stated otherwise, it should be assumed that the inputs are on the left side and the outputs on the right side. But it can be modified if this helps with the distribution of a scheme or to interpret the device.

Symbol	Description
	**Operator of functions matemáticas,símbolo** general. f(x1, ..., xn) must be replaced by a appropriate indication or a reference what characterize the function. x1, ..., xn must be replaced by one appropriate indication of the argument of the function.

<table>
<tr><td></td><td>For avoid all ambiguity with the converter symbols of level and converter of code no must to be employed the bar leaning to indicate the division.</td></tr>
<tr><td>
</td><td>Multiplier u=-2ab</td></tr>
<tr><td>
</td><td>Multiplier-divisor
u = ab/c</td></tr>
<tr><td>


</td><td>Amplificador,símbolo general.

They are also valid. two other symbols.

$U_i = m \cdot m_i \cdot f(w_1 a_1, w_2 ..., w_n a_n)$ Where $i = 1, 2, \ldots, k$

If an element performs a specific function besides of the amplification, 'f' can be replaced by a symbol badge appropriate. Another form "f" must be omitted.

If they will use the symbols distinguishers following for the indicated functions.

Σ sum
∫ integration</td></tr>
</table>

$\frac{d}{dt}$ derivative with respect to
of time
exponential function
logarithmic function
(base 10)

Sampling and retention

$m \cdot m_i$ is equal to the factor of amplification of the output I represent the factor common amplification

If the common factor is fixed y must to be represented, "m" must to be replaced by a number or an expression what gives the absolute value of the common factor or of the range within which it is set.

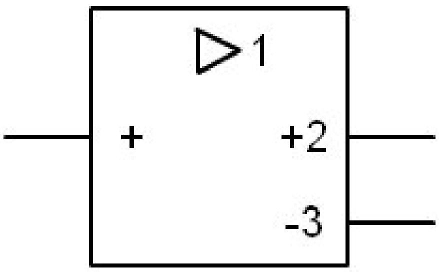
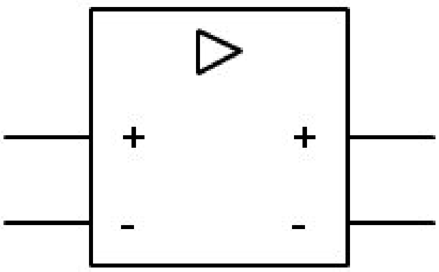
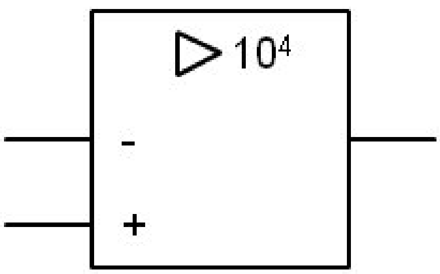
If the common factor is variable y is necessary show this must to preserve oneself the indication "m" and must indicate the method for determine its value, be it inside the symbol or in documentation of support.

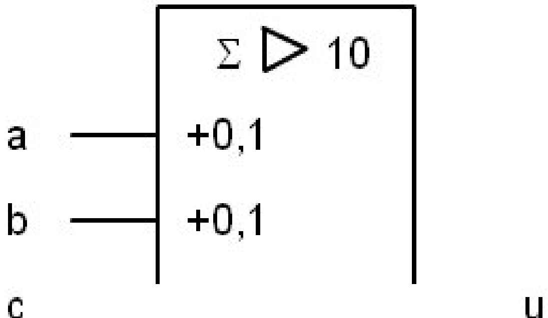
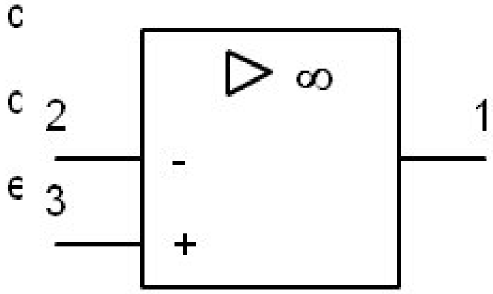
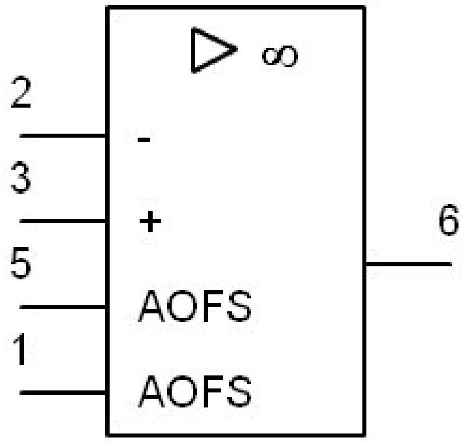
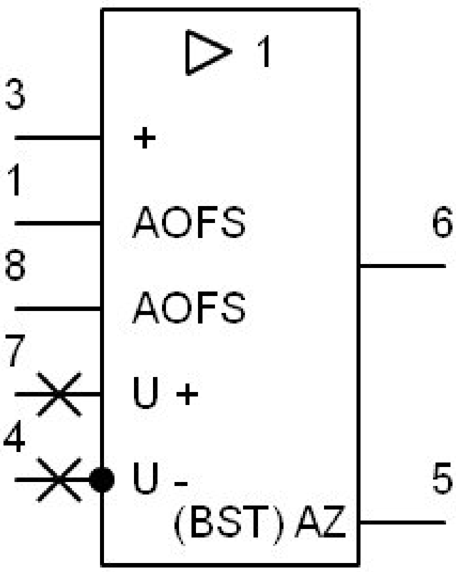
Otherwise the 'm' it should be omitted.

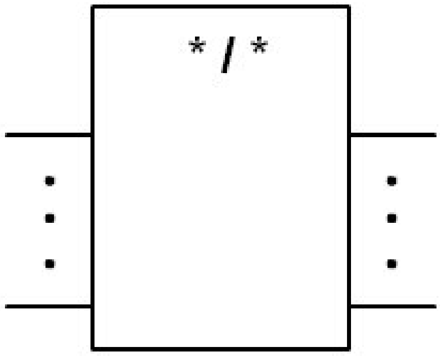
If they recommend the symbols following for the indication of the factor common

∞ if the common factor is large
if the common factor is 1
a number if the factor common must be indicated explicitly

***1...*2 if the common factor** it is fixed on the range *1...*2, *1...*2 must be replaced by the factor minimum and the factor maximum...m_k
they represent the values

	of amplification with its signs. If the factor of amplification is 1 the '1' may be omitted. If there is only one exit that is not specified in another way and if the amplification factor with its sign is equal to +1, the "+1" can be omitted w_1 ..., w_n represent the values of the weighting factors with its signs. If the value of the factor of the weight is 1, the '1' it can be omitted.
	**Amplifier with two** outputs, one of them direct with a amplification of 2, the another inverse with a amplification of -3
	**Amplification differential with two outputs** whose amplification is not specified
	**Amplifier differential of high profit, with** an amplification nominal of 10,000
	Summing amplifier, u = -10 (0.1a + 0.1b +

	0.2c + 0.5d + 1.0e) = -(a+b+2c+5d+10e)
	**Amplifier operational** Example: part of LM324
	**Amplifier operational** Example: LM741
	**Amplifier-follower of tension. Example:** LM310, enveloping metallic. The point represents the connection of the enveloping a a terminal.
	Converter, symbol



general

The distinctive symbol of operator */* can be replaced by *//* to indicate existence of a separation electric

The asterisks must to be replaced by appropriate indications for the quantities or implied qualities

The asterisk of the left refers to the entry, that of the right to the exit

It is advisable to use the following indications for the functions numbered

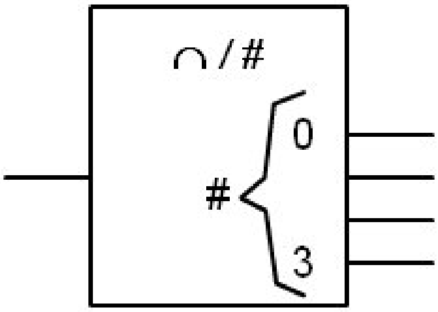
#digital, code no specified
∩ analog, function no specified
UoVtension
frequency
φ o Φ phase
Icurrent
Temperature

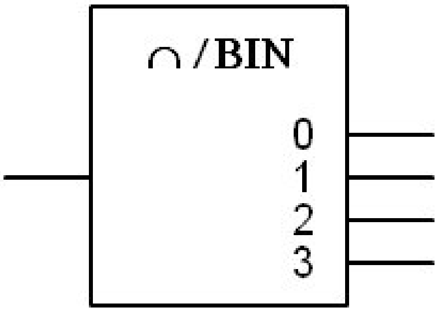
Notas:

1 The symbols distinctive generals of operator #/ ∩ y ∩/# they can to be replaced by DAC and ADC

2 In the symbols distinctions of the operators #/ ∩ y ∩/#, # can to be replaced by one appropriate indication of the code used in the entries digital [outputs] to determine represent the value internal. In this case, the inputs [outputs] digital must to be

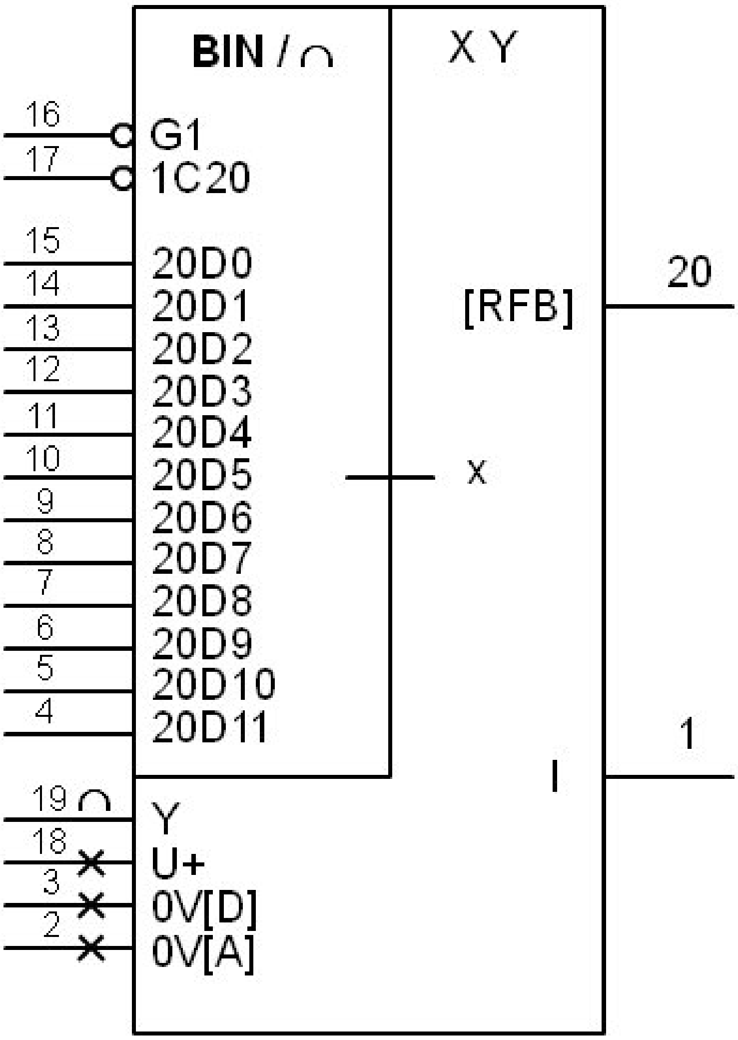
marked by characters that refer to this code.



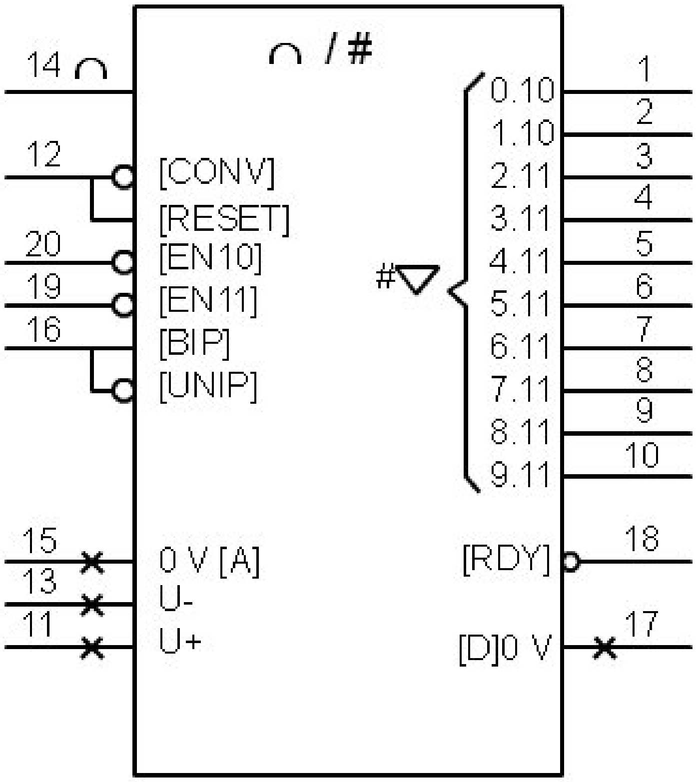
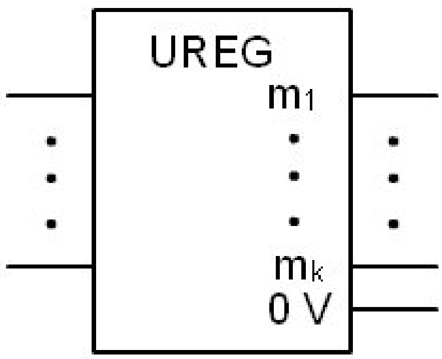
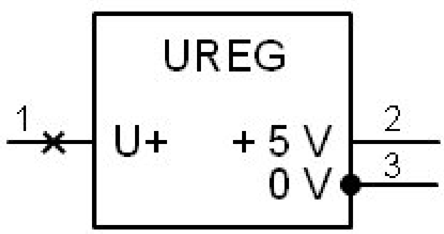


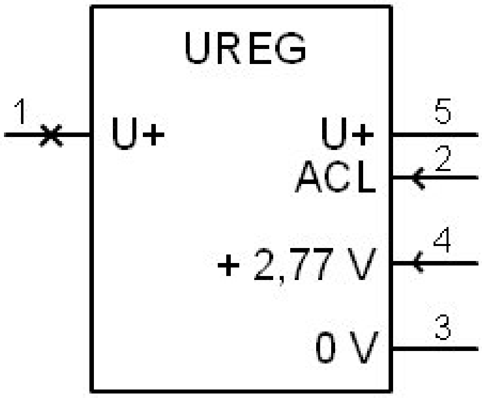
Converter analog/digital
convert the signal input analog a digital code weighted four elements binaries (bits)

Both symbols are valid



Converter digital analog (CDA) multiplier. Example: AD7545

	**Converter analog-digital** (CAD). Example: AD573
	**Voltage regulator.** General symbol m_1 ... m_k represent the tensions regulated (stabilized) with regarding to the terminal common (0 V) m_1 ... m_k must be replaced by: - U_1... U_k followed by each one by the sign of polarity or values real or ranges effective of the regulated tensions
	**Voltage regulator fixed value positive.** Example: LM309H
	Voltage regulator positive value of exit adjustable. Example: LM317T

<table>
<tr><td>


</td><td>

Voltage regulator positive adjustable with limitation of current. Example: L200CV

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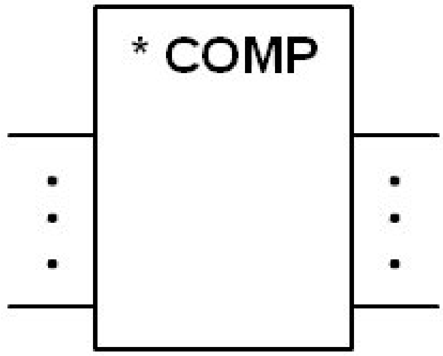

</td><td>

Comparator, symbol general

The asterisk must be replaced for the appropriate literal symbol para la magnitud o los operands whose values van a to compare oneself.

Can omit this literal symbol if not produce with Hello no confusion.

</td></tr>
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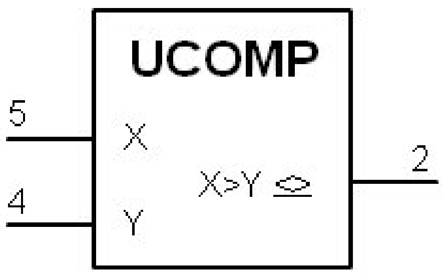

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Comparator of tensions. Example: part of LM339

</td></tr>
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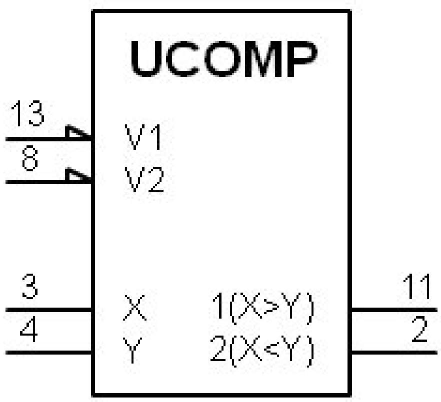

</td><td>

Comparator of tensions. Example: LM361

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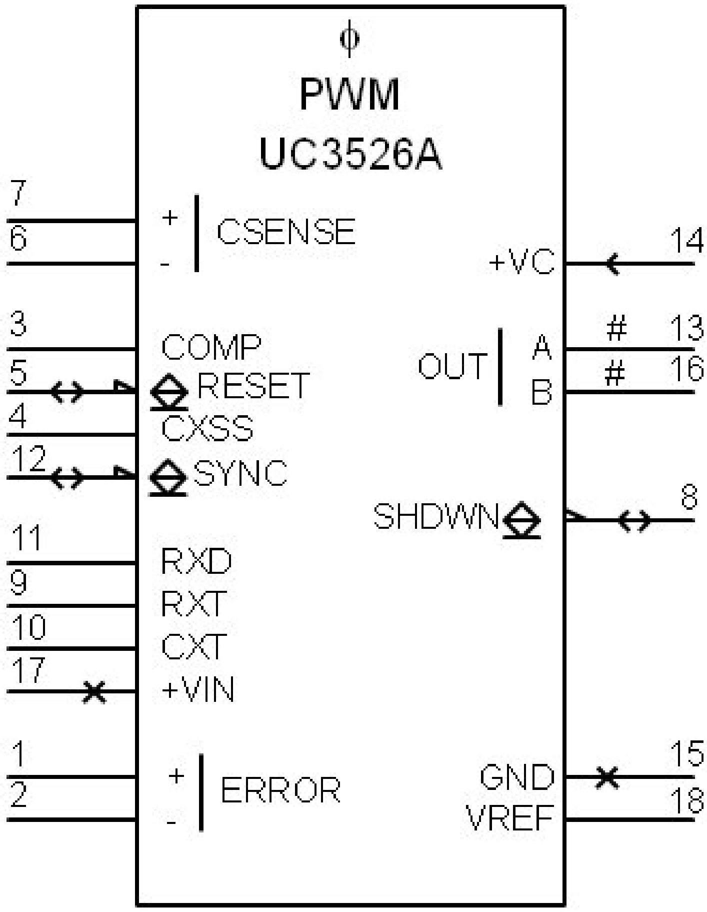
<tr><td>


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Width modulator of impulse. Example: Unitrode UC3526 A
</td></tr>
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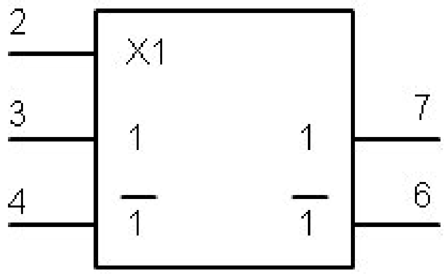

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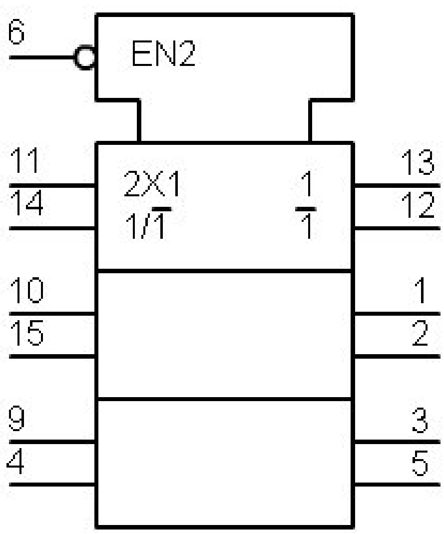
Switch analog electronic. Example: TL604
</td></tr>
<tr><td>


</td><td>

Multiplexer / Demultiplexer triple analog of two directions. Example: 74HC4053
</td></tr>
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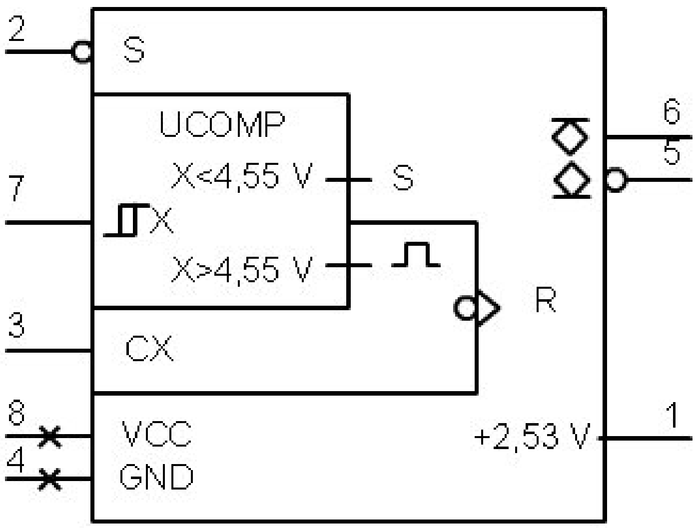	**Supervisor of tensions.** Example: TL7705 A

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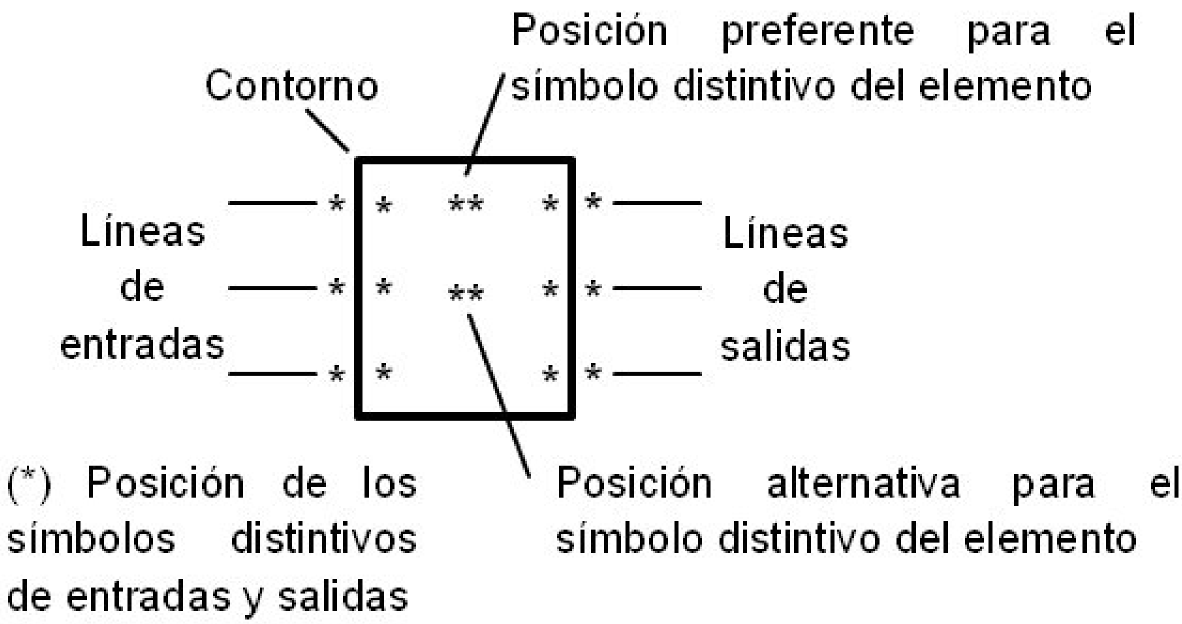
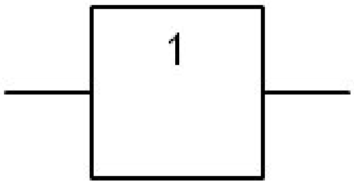
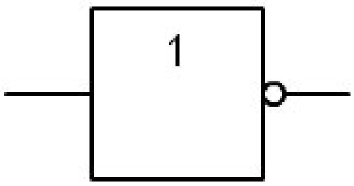
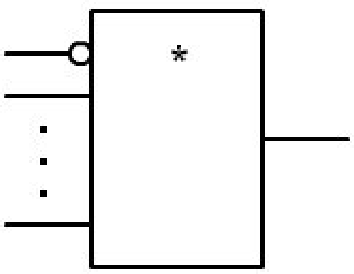
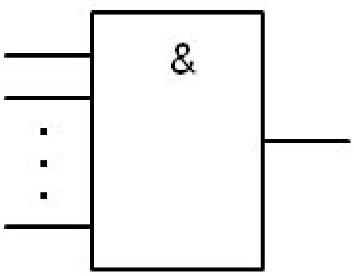
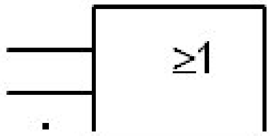
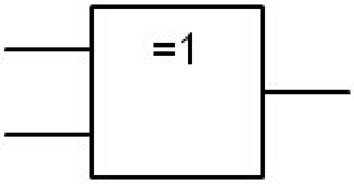
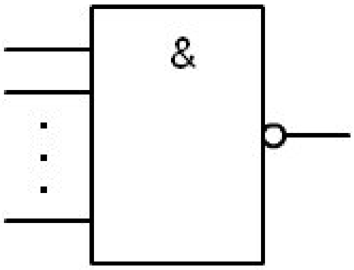
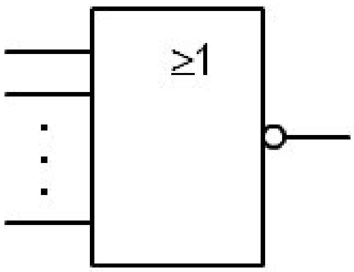
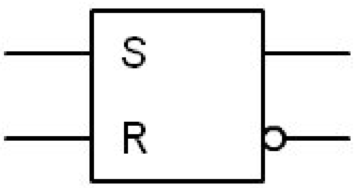
8.- Binary logical operators

The composition of this type of elements will be the same as that of the analog operators.



Symbol	Description
	**Logical IF gate (buffer)**
	**NOT gate or inverter (NOT)**
	**Logic gate with a negated input. (The circle** denies
	Logical AND gate. The output is 1 when all the tickets are 1.
	Logical OR gate. The output is 1 when

	Any of the entries is 1.
	**Exclusive OR (XOR) logic gate. The output is 1 if** only one entry is 1.
	**NAND logic gate. It is the negation of the** door Y.
	**NOR gate. It is the negation of the** door O.
	**Bistable R-S**

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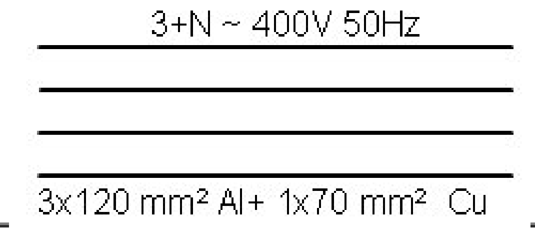
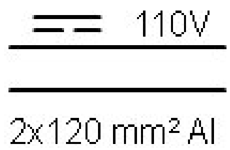
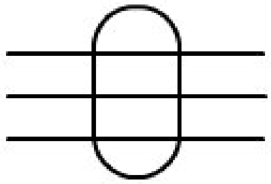
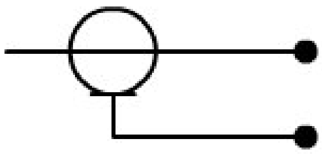
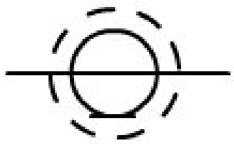
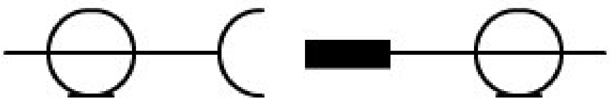
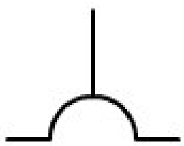
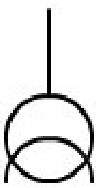
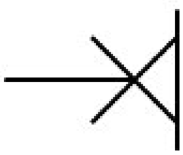
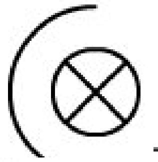
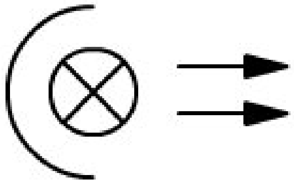
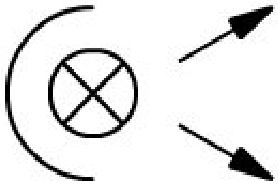
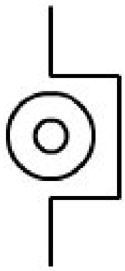
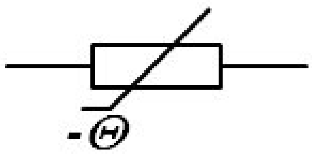
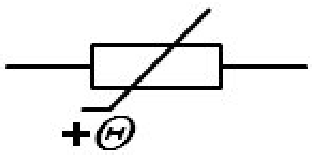
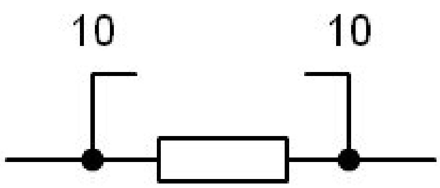
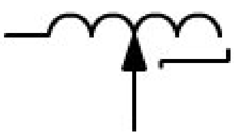
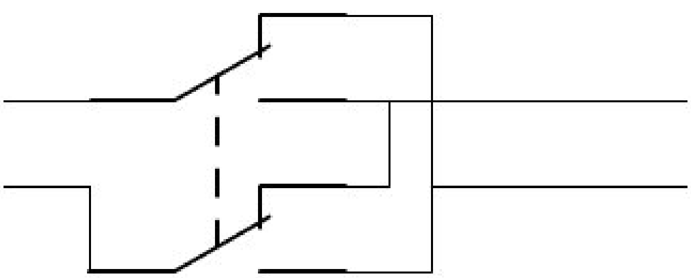
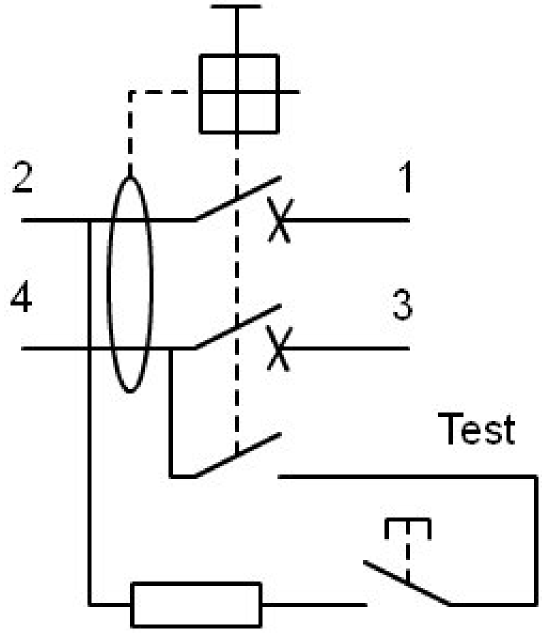
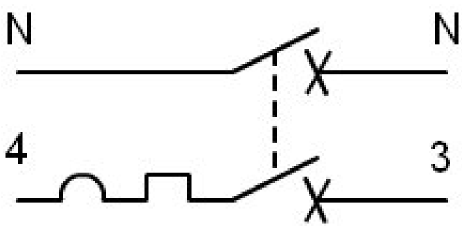
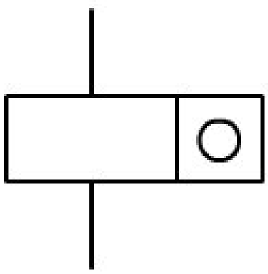
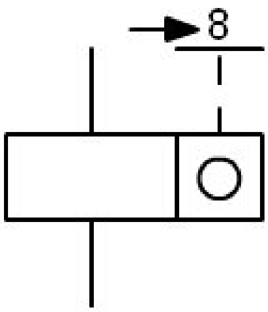
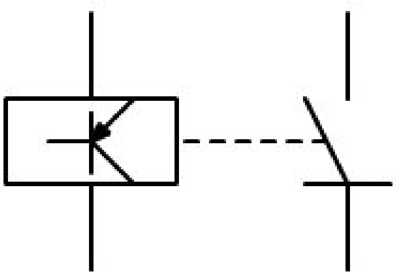
9.- Examples

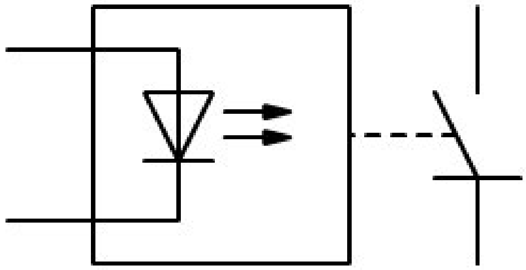
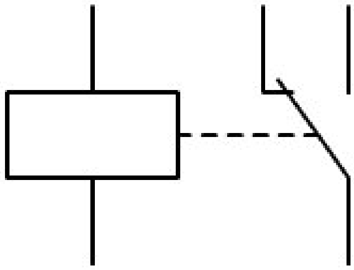
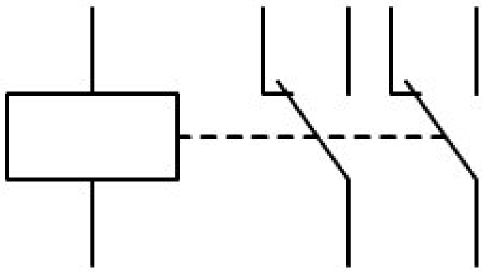
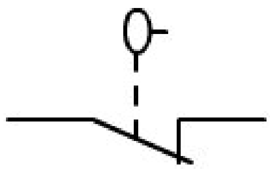
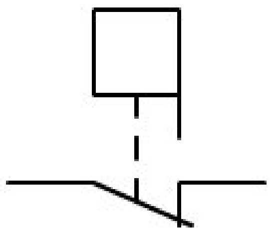
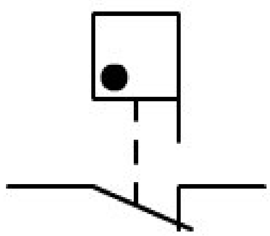
To obtain symbols that are not represented in the standard, they are obtained as combination of the previous ones, following the guidelines of that regulation. Below are some examples.

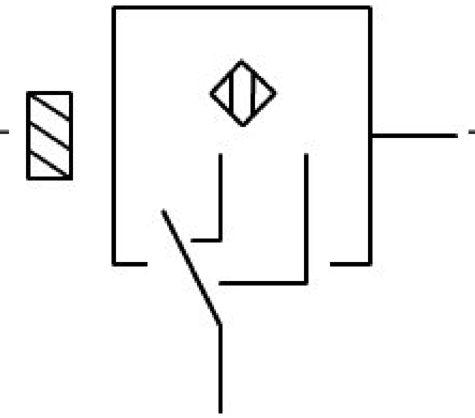
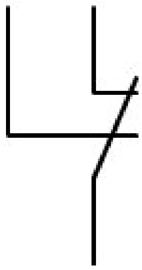
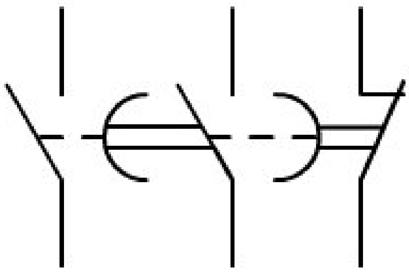
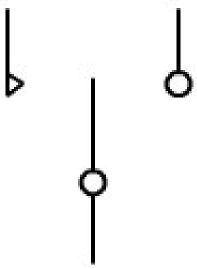
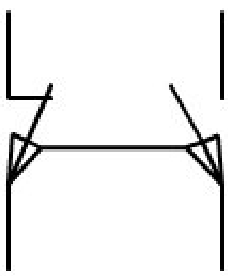
Symbol	Description
	Drivers of current circuit

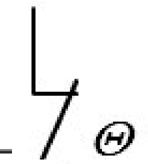
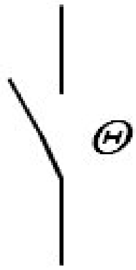
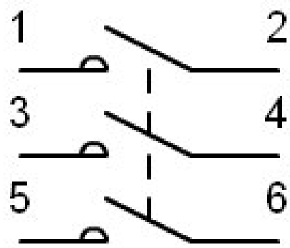
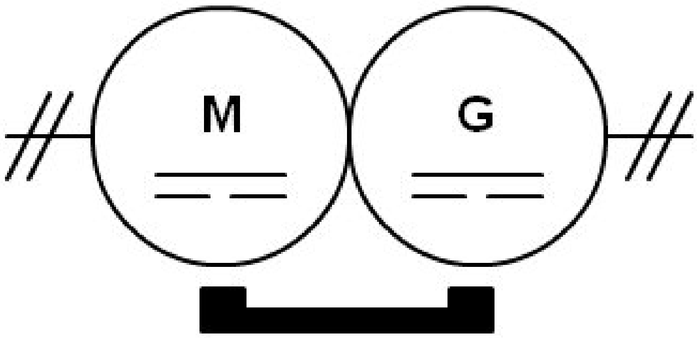
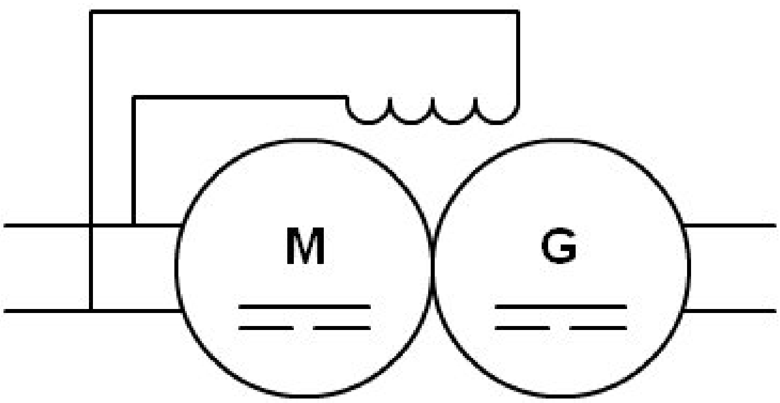
	three-phase, 400 V, 50 Hz, three conductors of 120 mm ^{2} of Aluminum, with thread neutral of 70 mm ^{2} of Copper.
	Drivers of current circuit continues, from 110 V, with two conductors 120 mm ^{2} of Aluminum.
	Drivers under a same cover o hose
	Coaxial cable with screen connected to terminals
	Cable coaxial screened
	Plug y base coaxial
	Plug base with shutter
	Base of plug (power) with transformer insulator. For example take for machine of to shave.
	Lighting intake on the wall. The channeling of connection comes through the left.

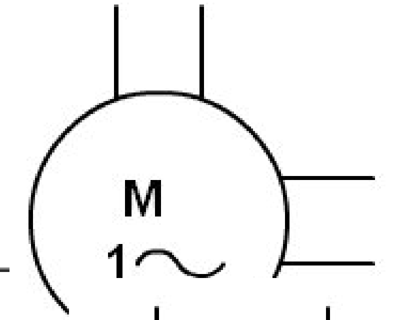
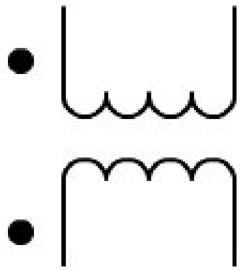
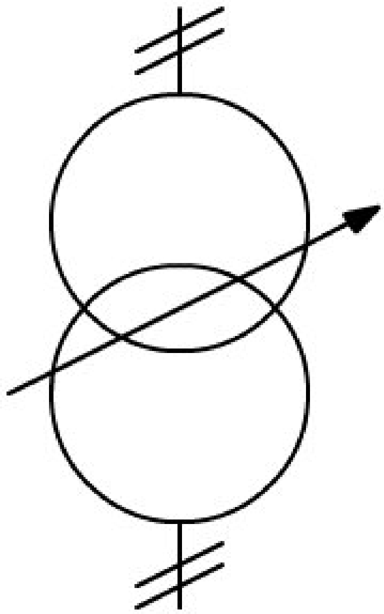
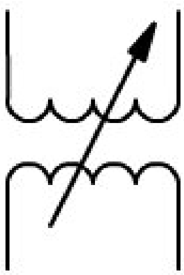
	Projector, symbol general
	Lighting projected
	Projector of lighting
	Button of pressure protected against functioning involuntary for middle of a cover glass protector that breaks.
	Resistance dependent on the temperature of shape negative (NTC)
	Resistance dependent on the temperature of shape positive (PTC)
	10 resistances parallel and identical
	Inductance with contact mobile variation for steps
	Circuit equivalent of the switch cross, represented in the norm as unifilar.

	Circuit breaker differential with test button. This it is a model of differential that is markets for the housing.
	Circuit breaker circuit breaker of one phase and neutral
	Pulse counter electric
	Accountant fixed manually a 8 impulses (setting to zero if 8 is replaced by 0)
	Electronic relay with closing contact semiconductor a thyristor base or triacs.

	Relay static driven by diode light emitter (optocoupler), with a closing contact base semiconductor of thyristors or triacs.
	Switch relay.
	Relay with double switch.
	Switch normally closed fluid level
	Switch normally closed flow rate of a fluid
	Switch normally closed gas flow rate
	Detector of capacitive proximity

	that works near a solid material
	Contact with two brands
	Contacts with two cuts
	Contact group with a contact of non-delayed closure a closing contact delayed when it is activate the device what contains the contact y a opening contact that is delayed when it is deactivated the device that contains the contact.
	Two-way contact with null position in the center and return automatic of a position (a the left) y sin automatic return in the opposite position.
	Switch of position operated mechanically in both directions with two circuits separated.
	Sensitive contact to the temperatura, contacto of closure. T can be

	replaced for conditions of temperature of operation.
	Sensitive contact to the temperatura, contacto of opening.
	Main contacts of power of a contactor with your numbering.
	Rotary converter, of direct current, with common excitement by permanent magnet
	Rotary converter, of direct current, with winding of common excitement

	Induction motor single-phase cage of squirrel, with the terminals del deprived auxiliary accessible
	Two transformer rollovers, the polarities of the tensions are indicated by points.
	Transformer with coupling regulable. Unifilar.
	Transformer with coupling adjustable. Multifilar

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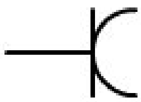
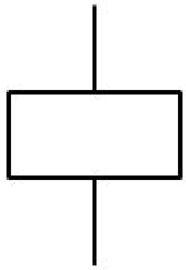
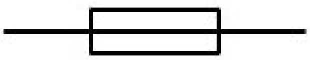
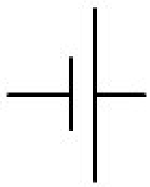
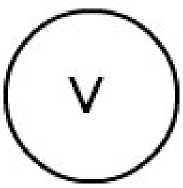
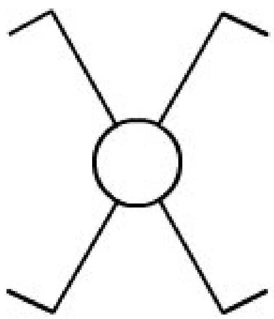
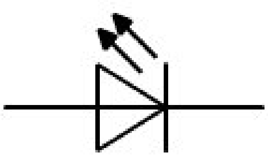
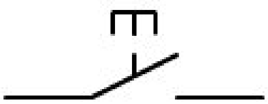
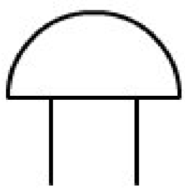
10.- Activities

1.- Draw the symbols in the corresponding gaps, both the multi-line and the single-line if it exists.

Switch	

DC motor	Lamp
Diode	Polarized capacitor

2.- Indicate the name of each of these symbols.

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